

Template

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1 写在前面

1.1 基础模版

```
1 #include <bits/stdc++.h>
2 using namespace std;
3 typedef long long ll;
4 #define REP( i, a, b ) for ( int i = ( a ); i <= ( b ); ++i )
5 #define REPD( i, a, b ) for ( int i = ( a ); i >= ( b ); --i )
6 inline ll rd() { /* {{{ */
7     ll r = 0, k = 1;
8     char c;
9     while ( !isdigit( c = getchar() ) )
10         if ( c == '-' )
11             k = -k;
12     while ( isdigit( c ) )
13         r = r * 10 + c - '0', c = getchar();
14     return r * k;
15 }
16 inline void prt( ll x ) {
17     printf( "%lld ", x );
18 } /* }}} */
19
20 int main() {
21
22     return 0;
23 }
```

1.2 vimrc

```
1 syntax on
2 set ts=2 sw=2 sts=2
3 set et ai cin nu sm ru cul
4 set mouse=a title wim=list nowrap ls=2
5 set ml sc
6 set mls=5
7 set fdm=marker
8 colo desert
9
10 im <C-S> <esc>:w<CR>
```

```

11 nn <C-S> :w<CR>
12 nn <space> za
13
14 " 使用 Windows 环境时可能需要
15 "     set shell=powershell
16 "     set backspace=indent,eol,start
17 "     set nocompatible
18 " set sh=powershell bs=indent,eol,start nosp
19 "
20 " 如果 <C-S> 导致 vim 卡死
21 " 在 .bashrc 里加一句 stty -ixon

```

2 数据结构

2.1 zkw 线段树

单点修区间查

```

1 ll s[N << 2], a[N];
2 int M;
3
4 ll f( ll x, ll y ) {
5     return x + y; // 改这
6 }
7
8 void build() {
9     for ( M = 1; M <= n + 1; M <= 1 ) {}
10    REP( i, 1, n ) s[i + M] = a[i];
11    REPD( i, M - 1, 1 ) s[i] = f( s[2 * i], s[2 * i + 1] );
12 }
13
14 ll qrang( int l, int r, ll init ) { // 根据 f 传 init
15     ll res = init;
16     for ( l = l + M - 1, r = r + M + 1; l ^ r ^ 1; l >>= 1, r >>= 1 )
17     {
18         if ( ~l & 1 )
19             res = f( res, s[l ^ 1] );
20         if ( r & 1 )
21             res = f( res, s[r ^ 1] );
22     }
23     return res;

```

```

23 }
24
25 void edit( int x, ll v ) {
26     for ( s[x += M] = v, x >>= 1; x; x >>= 1 ) {
27         s[x] = f( s[2 * x], s[2 * x + 1] );
28     }
29 }
30
31 ll qpoint( int x ) {
32     return s[x + M];
33 }

```

2.2 珂朵莉树

```

1 struct node {
2     int l, r;
3     mutable int v;
4     bool operator<( const node& rhs ) const { return l < rhs.l; }
5 };
6
7 set<node> odt;
8 typedef set<node>::iterator iter;
9
10 iter split( ll p ) {
11     iter tmp = odt.lower_bound( (node) { p, 0, 0 } );
12     if ( tmp != odt.end() && tmp->l == p )
13         return tmp;
14     --tmp;
15     int tl = tmp->l, tr = tmp->r, tv = tmp->v;
16     odt.erase( tmp );
17     odt.insert( (node) { tl, p - 1, tv } );
18     return odt.insert( (node) { p, tr, tv } ).first;
19 }
20
21 // 修改和查询注意 split 顺序
22 // iter itr=split(r+1), itl=split(l);

```

2.3 FHQ-Treap

以模版文艺平衡树为例

```

1  int n, m, clk, rt;
2  struct node {
3      int key, val, sz, tag, ls, rs;
4  } t[N];
5  int newnode( int k ) {
6      return t[++clk] = (node) { k, rand(), 1, 0 }, clk;
7  }
8  void down( int o ) {
9      if ( t[o].tag ) {
10         t[t[o].ls].tag = 1 - t[t[o].ls].tag;
11         t[t[o].rs].tag = 1 - t[t[o].rs].tag;
12         swap( t[t[o].ls].ls, t[t[o].ls].rs );
13         swap( t[t[o].rs].ls, t[t[o].rs].rs );
14         t[o].tag = 0;
15     }
16 }
17 void up( int o ) {
18     t[o].sz = t[t[o].ls].sz + t[t[o].rs].sz + 1;
19 }
20 void split( int o, int x, int& L, int& R ) {
21     if ( o == 0 )
22         return L = R = 0, void();
23     down( o );
24     if ( t[t[o].ls].sz + 1 >= x )
25         R = o, split( t[o].ls, x, L, t[o].ls );
26     else
27         L = o, split( t[o].rs, x - t[t[o].ls].sz - 1, t[o].rs, R );
28     up( o );
29 }
30 int merge( int L, int R ) {
31     if ( L == 0 || R == 0 )
32         return L + R;
33     if ( t[L].val > t[R].val )
34         return down( L ), t[L].rs = merge( t[L].rs, R ), up( L ), L;
35     else
36         return down( R ), t[R].ls = merge( L, t[R].ls ), up( R ), R;
37 }

```

2.4 并查集

```

1 ll n, fa[N];
2 void init() {
3     iota( fa + 1, fa + n + 1, 1 );
4 }
5
6 int find( int x ) {
7     if ( x == fa[x] )
8         return x;
9     return fa[x] = find( fa[x] );
10 }
11
12 void merge( int x, int y ) {
13     x = find( x ), y = find( y );
14     if ( x != y )
15         fa[x] = y;
16 }

```

2.5 ST 表

```

1 ll n, a[N], st[N][SP + 10], to[N][SP + 10], l2g[N];
2 ll op( ll x, ll y ) {
3     return max( x, y );
4 }
5
6 void init() {
7     l2g[1] = 0, to[n + 1][0] = n + 1;
8     REP( i, 2, n ) l2g[i] = l2g[i - 1] + !( i & ( i - 1 ) );
9     REP( i, 1, n ) st[i][0] = a[i], to[i][0] = i + 1;
10    REP( i, 1, SP ) {
11        REP( j, 1, n ) {
12            to[j][i] = to[to[j][i - 1]][i - 1];
13            st[j][i] = op( st[j][i - 1], st[to[j][i - 1]][i - 1] );
14        }
15    }
16 }
17
18 ll query( ll l, ll r ) { // [l, r] 闭区间
19     ++r;
20     ll d = l2g[r - l];
21     return op( st[l][d], st[r - ( 1 << d )][d] );

```

```
22 }
```

2.6 树状数组

```
1 ll n, fwt[N];
2 ll prod( ll x, ll d ) {
3     return x + d;
4 }
5 ll op( ll x, ll y ) {
6     return x + y;
7 }
8
9 void edit( int x, ll d ) {
10     for ( ; x <= n; x += x & -x )
11         fwt[x] = prod( fwt[x], d );
12 }
13
14 ll query( int x ) {
15     assert( 1 <= x && x <= n );
16     ll res = fwt[x];
17     x -= x & -x;
18     // 这种写法不用考虑最大或最小值的初值问题
19     for ( ; x; x -= x & -x )
20         res = op( res, fwt[x] );
21     return res;
22 }
```

2.7 线段树

2.7.1 我一般写的

```
1 ll n, a[N];
2 struct seg {
3     int l, r;
4     ll sum, tag;
5 } s[N << 2];
6 #define lson ( o * 2 )
7 #define rson ( o * 2 + 1 )
8
9 void down( int o ) {
```

```

10     if ( s[o].tag ) {
11         s[lson].sum += s[o].tag * ( s[lson].r - s[lson].l + 1 ),
12         s[lson].tag += s[o].tag;
13         s[rson].sum += s[o].tag * ( s[rson].r - s[rson].l + 1 ),
14         s[rson].tag += s[o].tag;
15         s[o].tag = 0;
16     }
17 }
18
19 void build( int o, int l, int r ) {
20     s[o].l = l, s[o].r = r;
21     if ( l == r )
22         return s[o].sum = a[l], void();
23     int mid = ( l + r ) / 2;
24     build( lson, l, mid );
25     build( rson, mid + 1, r );
26     s[o].sum = s[lson].sum + s[rson].sum;
27 }
28
29 void add( int o, int l, int r, ll d ) {
30     if ( l <= s[o].l && s[o].r <= r )
31         return s[o].sum += d * ( s[o].r - s[o].l + 1 ), s[o].tag += d,
32         void();
33     down( o );
34     int mid = ( s[o].l + s[o].r ) / 2;
35     if ( l <= mid )
36         add( lson, l, r, d );
37     if ( r > mid )
38         add( rson, l, r, d );
39     s[o].sum = s[lson].sum + s[rson].sum;
40 }
41
42 ll sum( int o, int l, int r ) {
43     if ( l <= s[o].l && s[o].r <= r )
44         return s[o].sum;
45     down( o );
46     int mid = ( s[o].l + s[o].r ) / 2;
47     ll res = 0;
48     if ( l <= mid )
49         res += sum( lson, l, r );

```

```

49     if ( r > mid )
50         res += sum( rson, l, r );
51     return res;
52 }

```

2.7.2 一款较通用的

```

1  ll n, a[N];
2
3  namespace sg {
4
5  struct Q { // 存修改, 存 tag
6      ll tag;
7      Q( ll tag = 0 ) : tag( tag ) {} // 注意初始值
8      void operator+=( const Q& q ) { tag += q.tag; }
9      // tag 的叠加
10 };
11 struct P { // 存记录维护的答案
12     ll sum;
13     P( ll sum = 0 ) : sum( sum ) {} // 注意初始值
14     void push( const Q& q, int l, int r ) { sum += q.tag * ( r - l +
15         1 ); }
16     // 区间从操作更新答案
17     void init( int x ) { sum = a[x]; } // 建树中的赋值
18 };
19 P operator&( const P& a, const P& b ) { // 子区间合并答案
20     return P( a.sum + b.sum );
21 }
22 // 基本上只需要修改上面的部分
23 P p[N << 2];
24 Q q[N << 2];
25 #define lson o * 2, l, ( l + r ) / 2
26 #define rson o * 2 + 1, ( l + r ) / 2 + 1, r
27 void up( int o, int l, int r ) {
28     if ( l < r )
29         p[o] = p[o * 2] & p[o * 2 + 1];
30 }
31 void push( const Q& v, int o, int l, int r ) {
32     q[o] += v;
33     p[o].push( v, l, r );

```

```

33 }
34 void down( int o, int l, int r ) {
35     push( q[o], lson );
36     push( q[o], rson );
37     q[o] = Q();
38 }
39 void build( int o = 1, int l = 1, int r = n ) {
40     if ( l == r )
41         p[o].init( l );
42     else {
43         build( lson );
44         build( rson );
45     }
46     up( o, l, r );
47 }
48 P query( int ql, int qr, int o = 1, int l = 1, int r = n ) {
49     if ( ql > r || l > qr )
50         return P();
51     if ( ql <= l && r <= qr )
52         return p[o];
53     down( o, l, r );
54     return query( ql, qr, lson ) & query( ql, qr, rson );
55 }
56 void update( int ql, int qr, const Q& v, int o = 1, int l = 1, int
    r = n ) {
57     if ( ql > r || l > qr )
58         return;
59     if ( ql <= l && r <= qr ) {
60         push( v, o, l, r );
61         return;
62     }
63     down( o, l, r );
64     update( ql, qr, v, lson );
65     update( ql, qr, v, rson );
66     up( o, l, r );
67 }
68
69 }
70 using namespace sg;

```

3 数学

3.1 快速幂

```
1 const ll MOD = 998244353; // 改模数
2
3 ll qpow( ll a, ll x ) {
4     ll res = 1;
5     a %= MOD;
6     while ( x ) {
7         if ( x & 1 )
8             res = res * a % MOD;
9         a = a * a % MOD, x >>= 1;
10    }
11    return res;
12 }
13
14 ll inv( ll x ) {
15     return qpow( x, MOD - 2 );
16 } // 模数为质数时
```

3.2 高斯消元

```
1 const int N = 110;
2 ll n;
3 double a[N][N], b[N];
4 void work() {
5     n = rd();
6     REP( i, 1, n ) {
7         REP( j, 1, n ) a[i][j] = rd();
8         b[i] = rd();
9     }
10    REP( i, 1, n ) {
11        int t = i;
12        REP( j, i + 1, n )
13            if ( abs( a[j][i] ) > 1e-7
14                && ( abs( a[t][i] ) > abs( a[j][i] ) || abs( a[t][i] ) < 1
15                    e-7 ) )
16            t = j;
17        REP( j, i, n ) swap( a[t][j], a[i][j] );
18    }
```

```

17     if ( abs( a[i][i] ) < 1e-7 ) {
18         puts( "No Solution" );
19         return 0;
20     }
21     swap( b[t], b[i] );
22     double e = a[i][i];
23     REP( j, i, n ) a[i][j] /= e;
24     b[i] /= e;
25     REP( j, i + 1, n ) {
26         double d = a[j][i];
27         REP( k, i, n ) a[j][k] -= d * a[i][k];
28         b[j] -= d * b[i];
29     }
30 }
31 REPd( i, n, 1 ) REP( j, 1, i - 1 ) b[j] -= a[j][i] * b[i], a[j][i]
    ] = 0;
32 // REP( i, 1, n ) printf( "%.2f\n", b[i] );
33 // b[1...n] 保存 Ax=b 的解
34 }

```

3.3 筛法

3.3.1 埃式筛

```

1  bitset<N> b;
2  ll n;
3  vector<ll> prime;
4  void erato() {
5      REP( i, 2, n ) if ( !b[i] ) {
6          prime.push_back( i );
7          for ( int j = i + i; j <= n; j += i )
8              b[j] = 1;
9      }
10 }

```

3.3.2 线性筛

```

1  bitset<N> b;
2  ll n, phi[N];
3  vector<ll> prime;
4  void euler() {

```

```

5  REP( i, 2, n ) {
6      if ( !b[i] ) {
7          prime.push_back( i );
8          phi[i] = i - 1;
9      }
10     for ( int p : prime ) {
11         if ( p * i > n )
12             break;
13         b[p * i] = 1;
14         if ( i % p == 0 ) {
15             phi[p * i] = phi[i] * p;
16             break;
17         }
18         phi[p * i] = phi[i] * phi[p];
19     }
20 }
21 }

```

3.4 类欧几里得

3.4.1 经典问题/我自己写的

```

1  ll floordiv( ll x, ll y ) {
2      assert( y != 0 );
3      if ( y < 0 )
4          x = -x;
5      return x >= 0 ? ( x / y ) : ( ( x - y + 1 ) / y );
6  }
7
8  ll mod( ll x, ll y ) {
9      return ( x % y + y ) % y;
10 }
11
12 ll f( ll a, ll b, ll c, ll n ) {
13     if ( c < 0 )
14         a = -a, b = -b, c = -c;
15     assert( c != 0 );
16     ll res = ( floordiv( a, c ) % MOD + MOD ) % MOD
17             * ( ( n * ( n + 1 ) / 2 ) % MOD ) % MOD
18             + ( floordiv( b, c ) % MOD + MOD ) % MOD * ( n + 1 ) %
19               MOD;

```

```

19     res %= MOD;
20     a = mod( a, c ), b = mod( b, c );
21     if ( !a )
22         return res;
23     res += ( n + 1 ) * ( floordiv( a * n + b, c ) % MOD ) % MOD
24         - floordiv( b, a ) % MOD;
25     res = ( res % MOD + MOD ) % MOD;
26     res -= f( c, a - b - 1, a, floordiv( a * n + b, c ) );
27     res = ( res % MOD + MOD ) % MOD;
28     return res;
29 }

```

3.4.2 洛谷模版题/OI-Wiki

$$f(a, b, c, n) = \sum_{i=0}^n \left\lfloor \frac{ai+b}{c} \right\rfloor$$

$$g(a, b, c, n) = \sum_{i=0}^n i \left\lfloor \frac{ai+b}{c} \right\rfloor$$

$$h(a, b, c, n) = \sum_{i=0}^n \left\lfloor \frac{an+b}{c} \right\rfloor^2$$

```

1 struct Data {
2     int f, g, h;
3     // f(a, b, c, n)=\sum_{i=0}^n \lfloor \frac{ai+b}{c} \rfloor
4     // g(a, b, c, n)=\sum_{i=0}^n i \lfloor \frac{ai+b}{c} \rfloor
5     // h(a, b, c, n)=\sum_{i=0}^n \lfloor \frac{an+b}{c} \rfloor^2
6 };
7
8 Data solve( long long a, long long b, long long c, long long n ) {
9     constexpr long long M = 998244353;
10    constexpr long long i2 = ( M + 1 ) / 2;
11    constexpr long long i6 = ( M + 1 ) / 6;
12    long long n2 = ( n + 1 ) * n % M * i2 % M;
13    long long n3 = ( 2 * n + 1 ) * ( n + 1 ) % M * n % M * i6 % M;
14    Data res = { 0, 0, 0 };
15    if ( a >= c || b >= c ) {
16        auto tmp = solve( a % c, b % c, c, n );
17        long long aa = a / c, bb = b / c;
18        res.f = ( tmp.f + aa * n2 + bb * ( n + 1 ) ) % M;
19        res.g = ( tmp.g + aa * n3 + bb * n2 ) % M;
20        res.h
21            = ( tmp.h + 2 * bb * tmp.f % M + 2 * aa * tmp.g % M + aa * aa
22                % M * n3 % M
23                + bb * bb % M * ( n + 1 ) % M + 2 * aa * bb % M * n2 % M

```

```

        )
23     % M;
24     return res;
25 }
26 long long m = ( a * n + b ) / c;
27 if ( !m )
28     return res;
29 auto tmp = solve( c, c - b - 1, a, m - 1 );
30 res.f = ( m * n - tmp.f + M ) % M;
31 res.g = ( m * n2 + ( M - tmp.f ) * i2 + ( M - tmp.h ) * i2 ) % M;
32 res.h = ( n * m % M * m - tmp.f - tmp.g * 2 + 3 * M ) % M;
33 return res;
34 }

```

3.5 递推组合数

```

1 ll C[N][N];
2 void calcC( int n ) {
3     C[0][0] = 1;
4     REP( i, 1, n ) {
5         C[i][0] = C[i][i] = 1;
6         REP( j, 1, i - 1 ) C[i][j] = ( C[i - 1][j] + C[i - 1][j - 1] )
            % MOD;
7     }
8 }

```

3.6 矩阵快速幂

```

1 struct mat {
2     ll n, m;
3     vector<vector<ll>> val; // 注意下标从 0 开始
4     mat( ll _n, ll _m ) : n( _n ), m( _m ) {
5         val.resize( _n );
6         REP( i, 0, _n - 1 ) val[i].resize( _m, 0 );
7     }
8     mat( ll _n, ll _m, vector<vector<ll>> _val )
9         : n( _n ), m( _m ), val( _val ) {}
10 };
11
12 mat mul( const mat& x, const mat& y, ll mod ) {

```

```

13  assert( x.m == y.n );
14  mat res( x.n, y.m );
15  REP( i, 0, res.n - 1 ) REP( j, 0, res.m - 1 ) {
16      REP( k, 0, x.m - 1 ) {
17          res.val[i][j] = ( res.val[i][j] + x.val[i][k] * y.val[k][j] %
18                          mod ) % mod;
19      }
20  }
21  return res;
22 }
23 mat qpow( mat a, ll x, ll mod ) {
24     assert( a.n == a.m );
25     mat res( a.n, a.n );
26     REP( i, 0, a.n - 1 ) res.val[i][i] = 1;
27     REP( i, 0, a.n - 1 ) REP( j, 0, a.m - 1 ) a.val[i][j] %= mod;
28     while ( x ) {
29         if ( x & 1 )
30             res = mul( res, a, mod );
31         a = mul( a, a, mod ), x >>= 1;
32     }
33     return res;
34 }

```

3.7 扩展欧几里得

```

1  ll exgcd( ll a, ll b, ll& x, ll& y ) {
2      if ( b == 0 ) {
3          x = 1, y = 0;
4          return a;
5      }
6      ll xx = 0, yy = 0;
7      ll res = exgcd( b, a % b, xx, yy );
8      y = xx - ( a / b ) * yy, x = yy;
9      return res;
10 }
11
12 ll inv( ll x, ll mod ) {
13     ll xx, yy;
14     ll d = exgcd( x, mod, xx, yy );

```

```

15     assert( d == 1 );
16     return ( xx % mod + mod ) % mod;
17 }

```

3.8 快速莫比乌斯/沃尔什变换

```

1  int n, m;
2  ll a[N], b[N], c[N];
3  ll inv2;
4
5  void solor() {
6      for ( int i = 0; i < n; ++i ) {
7          for ( int j = 0; j < m; ++j ) {
8              if ( ( j >> i ) & 1 )
9                  continue;
10             ( a[j + ( 1 << i )] += a[j] ) %= MOD;
11             ( b[j + ( 1 << i )] += b[j] ) %= MOD;
12         }
13     }
14     for ( int j = 0; j < m; ++j )
15         c[j] = ( a[j] * b[j] ) % MOD;
16     for ( int i = 0; i < n; ++i ) {
17         for ( int j = 0; j < m; ++j ) {
18             if ( ( j >> i ) & 1 )
19                 continue;
20             a[j + ( 1 << i )] = ( a[j + ( 1 << i )] - a[j] + MOD ) % MOD;
21             b[j + ( 1 << i )] = ( b[j + ( 1 << i )] - b[j] + MOD ) % MOD;
22             c[j + ( 1 << i )] = ( c[j + ( 1 << i )] - c[j] + MOD ) % MOD;
23         }
24     }
25     for ( int j = 0; j < m; ++j )
26         prt( c[j] );
27     puts( "" );
28 }
29
30 void soland() {
31     for ( int i = 0; i < n; ++i ) {
32         for ( int j = 0; j < m; ++j ) {
33             if ( ( j >> i ) & 1 )
34                 continue;

```

```

35     ( a[j] += a[j + ( 1 << i )] ) %= MOD;
36     ( b[j] += b[j + ( 1 << i )] ) %= MOD;
37 }
38 }
39 for ( int j = 0; j < m; ++j )
40     c[j] = ( a[j] * b[j] ) % MOD;
41 for ( int i = 0; i < n; ++i ) {
42     for ( int j = 0; j < m; ++j ) {
43         if ( ( j >> i ) & 1 )
44             continue;
45         a[j] = ( a[j] - a[j + ( 1 << i )] + MOD ) % MOD;
46         b[j] = ( b[j] - b[j + ( 1 << i )] + MOD ) % MOD;
47         c[j] = ( c[j] - c[j + ( 1 << i )] + MOD ) % MOD;
48     }
49 }
50 for ( int j = 0; j < m; ++j )
51     prt( c[j] );
52 puts( "" );
53 }
54
55 void solxor() {
56     for ( int i = 0; i < n; ++i ) {
57         for ( int j = 0; j < m; ++j ) {
58             if ( ( j >> i ) & 1 )
59                 continue;
60             ( a[j] += a[j + ( 1 << i )] ) %= MOD;
61             a[j + ( 1 << i )] = ( a[j] - 2 * a[j + ( 1 << i )] + 2 * MOD
62                 ) % MOD;
63             ( b[j] += b[j + ( 1 << i )] ) %= MOD;
64             b[j + ( 1 << i )] = ( b[j] - 2 * b[j + ( 1 << i )] + 2 * MOD
65                 ) % MOD;
66         }
67     }
68     for ( int j = 0; j < m; ++j )
69         c[j] = ( a[j] * b[j] ) % MOD;
70     for ( int i = 0; i < n; ++i ) {
71         for ( int j = 0; j < m; ++j ) {
72             if ( ( j >> i ) & 1 )
73                 continue;
74             a[j] = ( ( a[j] + a[j + ( 1 << i )] ) * inv2 ) % MOD;

```

```

73     a[j + ( 1 << i )] = ( a[j] - a[j + ( 1 << i )] + MOD ) % MOD;
74     b[j] = ( ( b[j] + b[j + ( 1 << i )] ) * inv2 ) % MOD;
75     b[j + ( 1 << i )] = ( b[j] - b[j + ( 1 << i )] + MOD ) % MOD;
76     c[j] = ( ( c[j] + c[j + ( 1 << i )] ) * inv2 ) % MOD;
77     c[j + ( 1 << i )] = ( c[j] - c[j + ( 1 << i )] + MOD ) % MOD;
78 }
79 }
80 for ( int j = 0; j < m; ++j )
81     prt( c[j] );
82 puts( "" );
83 }

```

3.9 子集卷积

```

1  const ll MOD = 1e9 + 9;
2  const int N = 1050576, SP = 25;
3  int n, m;
4  int popcnt[N];
5  ll a[SP][N], b[SP][N], c[SP][N];
6  ll inv2;
7
8  void sol() {
9      for ( int k = 0; k <= n; ++k ) {
10         for ( int i = 0; i < n; ++i ) {
11             for ( int j = 0; j < m; ++j ) {
12                 if ( ( j >> i ) & 1 )
13                     continue;
14                 ( a[k][j + ( 1 << i )] += a[k][j] ) %= MOD;
15                 ( b[k][j + ( 1 << i )] += b[k][j] ) %= MOD;
16             }
17         }
18     }
19     for ( int k = 0; k <= n; ++k ) {
20         for ( int l = 0; l <= k; ++l ) {
21             for ( int j = 0; j < m; ++j ) {
22                 c[k][j] = ( c[k][j] + a[l][j] * b[k - l][j] % MOD ) % MOD;
23             }
24         }
25     }
26     for ( int k = 0; k <= n; ++k ) {

```

```

27     for ( int i = 0; i < n; ++i ) {
28         for ( int j = 0; j < m; ++j ) {
29             if ( ( j >> i ) & 1 )
30                 continue;
31             c[k][j + ( 1 << i )] = ( c[k][j + ( 1 << i )] - c[k][j] +
                MOD ) % MOD;
32         }
33     }
34 }
35 for ( int j = 0; j < m; ++j )
36     prt( c[popcnt[j]][j] );
37 puts( "" );
38 }

```

4 图论

4.1 倍增

```

1 void dfs( int x, int fa ) {
2     pa[x][0] = fa;
3     dep[x] = dep[fa] + 1;
4     REP( i, 1, SP ) pa[x][i] = pa[pa[x][i - 1]][i - 1];
5     for ( int& v : g[x] )
6         if ( v != fa ) {
7             dfs( v, x );
8         }
9 }
10
11 int lca( int x, int y ) {
12     if ( dep[x] < dep[y] )
13         swap( x, y );
14     int t = dep[x] - dep[y];
15     REP( i, 0, SP ) if ( t & ( 1 << i ) ) x = pa[x][i];
16     REPd( i, SP - 1, -1 ) {
17         int xx = pa[x][i], yy = pa[y][i];
18         if ( xx != yy )
19             x = xx, y = yy;
20     }
21     return x == y ? x : pa[x][0];
22 }

```

4.2 网络流

不是我写的，但是看着还好

其中 11 是我改的，不敢保证有没有漏改，但是过了洛谷模版题

4.2.1 最大流

```
1  constexpr ll INF = LLONG_MAX / 2;
2
3  struct E {
4      int to;
5      ll cp;
6      E( int to, ll cp ) : to( to ), cp( cp ) {}
7  };
8
9  struct Dinic {
10     static const int M = 1E5 * 5;
11     int m, s, t;
12     vector<E> edges;
13     vector<int> G[M];
14     int d[M];
15     int cur[M];
16
17     void init( int n, int s, int t ) {
18         this->s = s;
19         this->t = t;
20         for ( int i = 0; i <= n; i++ )
21             G[i].clear();
22         edges.clear();
23         m = 0;
24     }
25
26     void addedge( int u, int v, ll cap ) {
27         edges.emplace_back( v, cap );
28         edges.emplace_back( u, 0 );
29         G[u].push_back( m++ );
30         G[v].push_back( m++ );
31     }
32
33     bool BFS() {
34         memset( d, 0, sizeof d );
```

```

35     queue<int> Q;
36     Q.push( s );
37     d[s] = 1;
38     while ( !Q.empty() ) {
39         int x = Q.front();
40         Q.pop();
41         for ( int& i : G[x] ) {
42             E& e = edges[i];
43             if ( !d[e.to] && e.cp > 0 ) {
44                 d[e.to] = d[x] + 1;
45                 Q.push( e.to );
46             }
47         }
48     }
49     return d[t];
50 }
51
52 ll DFS( int u, ll cp ) {
53     if ( u == t || !cp )
54         return cp;
55     ll tmp = cp, f;
56     for ( int& i = cur[u]; i < G[u].size(); i++ ) {
57         E& e = edges[G[u][i]];
58         if ( d[u] + 1 == d[e.to] ) {
59             f = DFS( e.to, min( cp, e.cp ) );
60             e.cp -= f;
61             edges[G[u][i] ^ 1].cp += f;
62             cp -= f;
63             if ( !cp )
64                 break;
65         }
66     }
67     return tmp - cp;
68 }
69
70 ll go() {
71     ll flow = 0;
72     while ( BFS() ) {
73         memset( cur, 0, sizeof cur );
74         flow += DFS( s, INF );

```

```

75     }
76     return flow;
77 }
78 } DC;

```

4.2.2 费用流

```

1  constexpr ll INF = LLONG_MAX / 2;
2
3  struct E {
4      int from, to;
5      ll cp, v;
6      E() {}
7      E( int f, int t, ll cp, ll v ) : from( f ), to( t ), cp( cp ), v(
        v ) {}
8  };
9
10 struct MCMF {
11     static const int M = 1E5 * 5;
12     int n, m, s, t;
13     vector<E> edges;
14     vector<int> G[M];
15     bool inq[M];
16     ll d[M], a[M];
17     int p[M];
18
19     void init( int _n, int _s, int _t ) {
20         n = _n;
21         s = _s;
22         t = _t;
23         REP( i, 0, n + 1 ) G[i].clear();
24         edges.clear();
25         m = 0;
26     }
27
28     void addedge( int from, int to, ll cap, ll cost ) {
29         edges.emplace_back( from, to, cap, cost );
30         edges.emplace_back( to, from, 0, -cost );
31         G[from].push_back( m++ );
32         G[to].push_back( m++ );

```

```

33     }
34
35     bool BellmanFord( ll& flow, ll& cost ) {
36         REP( i, 0, n + 1 ) d[i] = INF;
37         memset( inq, 0, sizeof inq );
38         d[s] = 0, a[s] = INF, inq[s] = true;
39         queue<int> Q;
40         Q.push( s );
41         while ( !Q.empty() ) {
42             int u = Q.front();
43             Q.pop();
44             inq[u] = false;
45             for ( int& idx : G[u] ) {
46                 E& e = edges[idx];
47                 if ( e.cp && d[e.to] > d[u] + e.v ) {
48                     d[e.to] = d[u] + e.v;
49                     p[e.to] = idx;
50                     a[e.to] = min( a[u], e.cp );
51                     if ( !inq[e.to] ) {
52                         Q.push( e.to );
53                         inq[e.to] = true;
54                     }
55                 }
56             }
57         }
58         if ( d[t] == INF )
59             return false;
60         flow += a[t];
61         cost += a[t] * d[t];
62         int u = t;
63         while ( u != s ) {
64             edges[p[u]].cp -= a[t];
65             edges[p[u] ^ 1].cp += a[t];
66             u = edges[p[u]].from;
67         }
68         return true;
69     }
70
71     pair<ll, ll> go() {
72         ll flow = 0, cost = 0;

```

```

73     while ( BellmanFord( flow, cost ) )
74         ;
75     return make_pair( flow, cost );
76 }
77 } MM;

```

4.3 二分图最大匹配

ps. 建单向图（即只有左部指向右部的边）

```

1  struct MaxMatch {
2      int n;
3      vector<int> G[N];
4      int vis[N], left[N], clk;
5
6      void init( int n ) {
7          this->n = n;
8          REP( i, 0, n + 1 ) G[i].clear();
9          memset( left, -1, sizeof left );
10         memset( vis, -1, sizeof vis );
11     }
12
13     bool dfs( int u ) {
14         for ( int v : G[u] )
15             if ( vis[v] != clk ) {
16                 vis[v] = clk;
17                 if ( left[v] == -1 || dfs( left[v] ) ) {
18                     left[v] = u;
19                     return true;
20                 }
21             }
22         return false;
23     }
24
25     int match() {
26         int ret = 0;
27         for ( clk = 0; clk <= n; ++clk )
28             if ( dfs( clk ) )
29                 ++ret;
30         return ret;
31     }

```

```
32 } MM;
```

4.4 Tarjan 强连通分量缩点

```
1 int low[N], dfn[N], clk, B, bl[N];
2 vector<int> bcc[N];
3 void init() {
4     B = clk = 0;
5     memset( dfn, 0, sizeof dfn );
6 }
7 void tarjan( int u ) {
8     static int st[N], p;
9     static bool in[N];
10    dfn[u] = low[u] = ++clk;
11    st[p++] = u;
12    in[u] = true;
13    for ( int& v : G[u] ) {
14        if ( !dfn[v] ) {
15            tarjan( v );
16            low[u] = min( low[u], low[v] );
17        } else if ( in[v] )
18            low[u] = min( low[u], dfn[v] );
19    }
20    if ( dfn[u] == low[u] ) {
21        ++B;
22        while ( 1 ) {
23            int x = st[--p];
24            in[x] = false;
25            bl[x] = B;
26            bcc[B].push_back( x );
27            if ( x == u )
28                break;
29        }
30    }
31 }
```

4.5 树直径

```
1 ll n, dep[N], mxdep[N];
2 vector<int> g[N];
```

```

3 vector<int> dmt;
4 void dfs1( int x, int fa ) {
5     dep[x] = dep[fa] + 1;
6     mxdep[x] = 1;
7     for ( int u : g[x] )
8         if ( u != fa ) {
9             dfs1( u, x );
10            mxdep[x] = max( mxdep[u] + 1, mxdep[x] );
11        }
12 }
13
14 void dfs2( int x, int fa ) { // 找一条直径，如果只需要直径长度则不
    用
15     dmt.push_back( x );
16     int nxt = -1;
17     for ( int u : g[x] )
18         if ( u != fa ) {
19             if ( nxt == -1 || mxdep[u] > mxdep[nxt] )
20                 nxt = u;
21         }
22     if ( nxt != -1 )
23         dfs2( nxt, x );
24 }
25
26 void diameter() {
27     dep[0] = 0;
28     dfs1( 1, 0 );
29     int rt = max_element( dep + 1, dep + n + 1 ) - dep;
30     dfs1( rt, 0 );
31     dfs2( rt, 0 );
32 }

```

4.6 树重心

```

1 ll n, sz[N], mxsz[N], G;
2 vector<int> g[N];
3 void dfs( int x, int fa ) {
4     sz[x] = 1, mxsz[x] = 0;
5     for ( int u : g[x] )
6         if ( u != fa ) {

```

```

7     dfs( u, x );
8     sz[x] += sz[u];
9     mxsz[x] = max( sz[u], mxsz[x] );
10    }
11    mxsz[x] = max( mxsz[x], n - sz[x] );
12    if ( G == -1 || ( mxsz[x] < mxsz[G] || ( mxsz[x] == mxsz[G] && x
        < G ) ) )
13        G = x;
14 }

```

4.7 树链剖分

```

1  ll fa[N], dep[N], son[N], sz[N], top[N], idx[N], clk, ridx[N], out[
    N];
2  vector<int> g[N];
3  void predfs( int x, int f ) { // 第一次 dfs 计算父亲/深度/大小/重儿
    子
4      fa[x] = f, dep[x] = dep[f] + 1, sz[x] = 1;
5      for ( auto v : g[x] )
6          if ( v != f ) {
7              predfs( v, x );
8              if ( sz[v] > sz[son[x]] )
9                  son[x] = v;
10                 sz[x] += sz[v];
11             }
12 }
13
14 void dfs( int x, int tp ) { // 第二次 dfs
15     // top 链的顶端
16     // idx 节点在 dfs 序中的编号
17     // ridx dfs 序中编号对应的节点
18     // out 以该节点为根的子树的编号的最大值
19     // 即 dfs 序中 [idx, out] 表示以它为根的子树
20     // [idx[top[x]], idx[x]] 表示这条链的顶端到该节点的一段链
21     top[x] = tp, idx[x] = ++clk, ridx[clk] = x;
22     if ( son[x] != 0 )
23         dfs( son[x], tp );
24     for ( auto v : g[x] )
25         if ( v != fa[x] && v != son[x] )
26             top[v] = v, dfs( v, v );

```

```

27     out[x] = clk;
28 }
29
30 void addpath( int u, int v, ll d ) {
31     int uu = top[u], vv = top[v];
32     while ( uu != vv ) {
33         if ( dep[uu] < dep[vv] )
34             swap( u, v ), swap( uu, vv );
35         add( 1, idx[uu], idx[u], d );
36         u = fa[uu], uu = top[u];
37     }
38     if ( dep[u] < dep[v] )
39         swap( u, v );
40     add( 1, idx[v], idx[u], d );
41 }
42
43 void addtree( int x, ll d ) {
44     add( 1, idx[x], out[x], d );
45 }
46
47 ll sumpath( int u, int v ) {
48     ll res = 0;
49     int uu = top[u], vv = top[v];
50     while ( uu != vv ) {
51         if ( dep[uu] < dep[vv] )
52             swap( u, v ), swap( uu, vv );
53         res = ( res + sum( 1, idx[uu], idx[u] ) ) % P;
54         u = fa[uu], uu = top[u];
55     }
56     if ( dep[u] < dep[v] )
57         swap( u, v );
58     res = ( res + sum( 1, idx[v], idx[u] ) ) % P;
59     return res;
60 }
61
62 ll sumtree( int x ) {
63     return sum( 1, idx[x], out[x] );
64 }

```

4.8 最短路

4.8.1 Floyd (最小环)

```
1 ll n, m, mincycle;
2 ll g[N][N], dis[N][N];
3 void floyd() {
4     // 如果 g[i][j] 之间没边则存 inf
5     // 注意 inf 的三倍不能爆 long long
6     mincycle = LLONG_MAX / 4;
7     REP( i, 1, n ) REP( j, 1, n ) dis[i][j] = g[i][j];
8     REP( k, 1, n ) {
9         REP( i, 1, k - 1 )
10            REP( j, i + 1, k - 1 ) {
11                mincycle = min( mincycle, dis[i][j] + g[j][k] + g[k][i] );
12            }
13         REP( i, 1, n )
14            REP( j, 1, n )
15                dis[i][j] = min( dis[i][j], dis[i][k] + dis[k][j] );
16     }
17 }
```

4.8.2 Spfa (判负环)

```
1 ll n, m, dis[N], inq[N], cnt[N];
2 vector<pair<int, ll>> g[N];
3 bool spfa( int s ) {
4     // 如果有负环则 return true
5     queue<int> q;
6     fill_n( dis + 1, n, LLONG_MAX / 2 );
7     fill_n( inq + 1, n, 0 );
8     fill_n( cnt + 1, n, 0 );
9     dis[s] = 0, inq[s] = cnt[s] = 1;
10    q.push( s );
11    while ( !q.empty() ) {
12        int u = q.front();
13        q.pop();
14        inq[u] = 0;
15        for ( auto [v, w] : g[u] ) {
16            if ( dis[v] > dis[u] + w ) {
17                dis[v] = dis[u] + w;
```

```

18         cnt[v] = cnt[u] + 1;
19         if ( cnt[v] > n )
20             return true;
21         if ( !inq[v] ) {
22             inq[v] = 1, q.push( v );
23         }
24     }
25 }
26 }
27 return false;
28 }

```

4.8.3 Dijkstra

```

1 ll n, m, dis[N], vis[N];
2 vector<pair<int, ll>> g[N];
3 void dijkstra( int s ) {
4     fill_n( dis + 1, n, LLONG_MAX / 2 );
5     fill_n( vis + 1, n, 0 );
6     dis[s] = 0;
7     priority_queue<pair<ll, int>> q;
8     q.push( make_pair( 0, s ) );
9     while ( !q.empty() ) {
10         int u = q.top().second;
11         q.pop();
12         if ( vis[u] )
13             continue;
14         vis[u] = 1;
15         for ( auto [v, w] : g[u] ) {
16             if ( vis[v] )
17                 continue;
18             if ( dis[v] > dis[u] + w ) {
19                 dis[v] = dis[u] + w;
20                 q.push( make_pair( -dis[v], v ) );
21             }
22         }
23     }
24 }

```

4.9 拓扑排序

```
1 ll n, in[N];
2 vector<int> g[N];
3 vector<int> res;
4 void topo() {
5     // 不要忘记在建图时记录入度
6     queue<int> q;
7     REP( i, 1, n )
8         if ( in[i] == 0 )
9             q.push( i );
10    while ( !q.empty() ) {
11        int u = q.front();
12        q.pop();
13        res.push_back( u );
14        for ( int v : g[u] ) {
15            --in[v];
16            if ( in[v] == 0 )
17                q.push( v );
18        }
19    }
20 }
```

4.10 最小生成树

Kruskal 算法

```
1 ll n, m, fa[N], weight_sum;
2 pair<ll, pair<int, int>> edge[M];
3 int find( int x ) {
4     if ( fa[x] == x )
5         return x;
6     return fa[x] = find( fa[x] );
7 }
8
9 void kruskal() {
10    weight_sum = 0;
11    sort( edge + 1, edge + m + 1 );
12    iota( fa + 1, fa + n + 1, 1 );
13    REP( i, 1, m ) {
14        auto [u, v] = edge[i].second;
```

```

15     u = find( u ), v = find( v );
16     if ( u == v )
17         continue;
18     fa[u] = v;
19     weight_sum += edge[i].first;
20 }
21 }

```

4.11 欧拉路径/回路

```

1 ll n, m, cnt[N];
2 vector<int> g[N];
3 vector<int> path;
4 void dfs( int x ) {
5     while ( cnt[x] < g[x].size() ) {
6         int u = g[x][cnt[x]];
7         ++cnt[x];
8         dfs( u );
9     }
10    path.push_back( x );
11    // 最后要 reverse(path.begin(), path.end());
12 }

```

5 字符串

5.1 KMP

```

1 char s1[N];
2 int n, pi[N];
3 void solve() {
4     // 字符串存于 s1[0, n)
5     pi[0] = 0;
6     REP( i, 1, n - 1 ) {
7         int j = pi[i - 1];
8         while ( j > 0 && s1[j] != s1[i] )
9             j = pi[j - 1];
10        if ( s1[j] == s1[i] )
11            ++j;
12        pi[i] = j;
13    }

```

```
14 }
```

5.2 Trie 树

```
1 int nxt[N][150], cnt;
2 int num[N];
3 void insert( char* s, int len ) {
4     // 字符串 s[0, len)
5     // 0 为根结点
6     int p = 0;
7     REP( i, 0, len - 1 ) {
8         if ( !nxt[p][s[i]] )
9             nxt[p][s[i]] = ++cnt;
10        p = nxt[p][s[i]];
11        ++num[p]; // 对前缀的统计在此处进行
12    }
13    // 对字符串整串的统计在此处进行
14 }
15
16 int find( char* s, int len ) {
17     int p = 0;
18     REP( i, 0, len - 1 ) {
19         if ( !nxt[p][s[i]] )
20             return 0;
21         p = nxt[p][s[i]];
22     }
23     return num[p];
24 }
```

5.3 Manacher

```
1 // s[0, n)
2 // d1[i] 表示 [i-d1[i], i+d1[i]] 是奇数长度回文 (i 为中心)
3 // d2[i] 表示 [i-d2[i]+1, i+d2[i]] 是偶数长度回文 (i+0.5 为中心)
4 // d2[i] 为 0 表示不存在
5
6 int l = 0, r = -1;
7 for ( int i = 0; i < n; ++i ) {
8     if ( r < i ) {
9         int j = i;
```

```

10     while ( j > 0 && 2 * i - ( j - 1 ) < n && s[j - 1] == s[2 * i -
11           ( j - 1 )] )
12         --j;
13         d1[i] = i - j;
14         l = j, r = 2 * i - j;
15     } else {
16         int j = ( l + r ) - i;
17         if ( j - d1[j] <= 1 ) {
18             int k = i * 2 - r;
19             while ( k > 0 && 2 * i - ( k - 1 ) < n
20                   && s[k - 1] == s[2 * i - ( k - 1 )] )
21                 --k;
22             d1[i] = i - k;
23             l = k, r = 2 * i - k;
24         } else
25             d1[i] = d1[j];
26     }
27
28     l = 0, r = -1;
29     for ( int i = 0; i + 1 < n; ++i ) {
30         if ( r < i ) {
31             int j = i;
32             while ( j >= 0 && i + i + 1 - j < n && s[j] == s[i + i + 1 - j]
33                   )
34                 --j;
35             d2[i] = i - j;
36             l = j + 1, r = i + i + 1 - j - 1;
37         } else {
38             int j = ( l + r ) - i - 1;
39             if ( j < 0 || j - d2[j] < 1 ) {
40                 int k = i + i + 1 - r - 1;
41                 while ( k >= 0 && i + i + 1 - k < n && s[k] == s[i + i + 1 -
42                       k] )
43                     --k;
44                 d2[i] = i - k;
45                 l = k + 1, r = i + i + 1 - k - 1;
46             } else
47                 d2[i] = d2[j];
48         }
49     }

```

```
47 | }
```

上面的太丑了下面另一版

```
1 | int RL[N];
2 | // 最后结果的 RL 的说明:
3 | // 加了字符 '#' 的字符串中 [i - RL[i], i + RL[i]] 是极大回文串
4 | // 在原串中 RL[i] 正好是以 (i-1)/2 为中心极大回文串长度 (不论奇偶)
5 | void manacher( char* a, int n ) { // "abc" => "#a#b#c#"
6 |     int r = 0, p = 0;
7 |     REP( i, 0, n - 1 ) {
8 |         if ( i < r )
9 |             RL[i] = min( RL[2 * p - i], r - i );
10 |         // 上面这行 r-i 还是 r-i+1 结果一样, 尊重原作者
11 |         else
12 |             RL[i] = 1;
13 |         while ( i - RL[i] >= 0 && i + RL[i] < n && a[i - RL[i]] == a[i
14 |             + RL[i]] )
15 |             RL[i]++;
16 |         if ( RL[i] + i - 1 > r ) {
17 |             r = RL[i] + i - 1;
18 |             p = i;
19 |         }
20 |     }
21 |     REP( i, 0, n - 1 ) --RL[i];
22 | }
```

5.4 回文自动机

```
1 | // num 是该结点表示的前缀的回文后缀个数
2 | // cnt 是该结点表示的回文串在原串中的出现次数 (使用前需要向父亲更
3 |     新)
4 | namespace pam {
5 |
6 | int t[N][26], fa[N], len[N], rs[N], cnt[N], num[N];
7 | int sz, n, last;
8 | int _new( int l ) {
9 |     len[sz] = l;
10 |     cnt[sz] = num[sz] = 0;
11 |     return sz++;
12 | }
```

```

12 }
13 // 记得 init
14 void init() {
15     memset( t, 0, sz * sizeof t[0] );
16     rs[n = sz = 0] = -1;
17     last = _new( 0 );
18     fa[last] = _new( -1 );
19 }
20 int get_fa( int x ) {
21     while ( rs[n - 1 - len[x]] != rs[n] )
22         x = fa[x];
23     return x;
24 }
25 void ins( int ch ) {
26     rs[++n] = ch;
27     int p = get_fa( last );
28     if ( !t[p][ch] ) {
29         int np = _new( len[p] + 2 );
30         num[np] = num[fa[np] = t[get_fa( fa[p] )][ch]] + 1;
31         t[p][ch] = np;
32     }
33     ++cnt[last = t[p][ch]];
34 }
35
36 }

```

5.5 字符串哈希

```

1 #define ENABLE_DOUBLE_HASH
2
3 namespace hashstr {
4
5     typedef unsigned long long ull;
6     const int x = 135;
7     const int p1 = 0x3b3b3bbb, p2 = 1e9 + 1011;
8     ull xp1[N], xp2[N];
9
10    // 记得 init
11    void init_xp() {
12        xp1[0] = xp2[0] = 1;

```

```

13     for ( int i = 1; i < N; ++i ) {
14         xp1[i] = xp1[i - 1] * x % p1;
15         xp2[i] = xp2[i - 1] * x % p2;
16     }
17 }
18
19 struct str {
20     char s[N];
21     int length;
22     ull h[N];
23
24     ull hash() {
25         length = strlen( s );
26         ull res1 = 0, res2 = 0;
27         h[length] = 0;
28         for ( int j = length - 1; j >= 0; --j ) {
29 #ifdef ENABLE_DOUBLE_HASH
30             res1 = ( res1 * x + s[j] ) % p1;
31             res2 = ( res2 * x + s[j] ) % p2;
32             h[j] = ( res1 << 32 ) | res2;
33 #else
34             res1 = ( res1 * x + s[j] ) % p1;
35             h[j] = res1;
36 #endif
37         }
38         return h[0];
39     }
40     // 左闭右开区间
41     ull get_substr_hash( int left, int right ) const {
42         int len = right - left;
43 #ifdef ENABLE_DOUBLE_HASH
44         unsigned int mask32 = ~( 0u );
45         ull left1 = h[left] >> 32, right1 = h[right] >> 32;
46         ull left2 = h[left] & mask32, right2 = h[right] & mask32;
47         return ( ( ( left1 - right1 * xp1[len] % p1 + p1 ) % p1 ) << 32
48                 | ( ( ( left2 - right2 * xp2[len] % p2 + p2 ) % p2 ) ) );
49 #else
50         return ( ( h[left] - h[right] * xp1[len] % p1 + p1 ) % p1 );
51 #endif

```

```

52     }
53 };
54
55 }

```

5.6 后缀数组

```

1  int buf1[N], buf2[N], buc[N];
2
3  // str 存储在下标 [1, n] 的范围内, str[0] 要与 str[1..n] 都不相同
4  // ht[i] = LCP(sa[i], sa[i-1])
5  void sufsort( char str[], int n, int sa[], int rk[], int ht[] ) {
6      int *x = buf1, *y = buf2, m = 127;
7      REP( i, 0, m ) buc[i] = 0;
8      REP( i, 1, n ) buc[x[i]] = str[i]++;
9      REP( i, 1, m ) buc[i] += buc[i - 1];
10     REPd( i, n, 1 ) sa[buc[x[i]]--] = i;
11     for ( int k = 1; k <= n; k <= 1 ) {
12         int p = 0;
13         REP( i, n - k + 1, n ) y[++p] = i;
14         REP( i, 1, n ) if ( sa[i] > k ) y[++p] = sa[i] - k;
15         REP( i, 0, m ) buc[i] = 0;
16         REP( i, 1, n ) buc[x[y[i]]]++;
17         REP( i, 1, m ) buc[i] += buc[i - 1];
18         REPd( i, n, 1 ) sa[buc[x[y[i]]]--] = y[i];
19         swap( x, y ), x[sa[1]] = p = 1;
20         REP( i, 2, n )
21             if ( y[sa[i - 1]] == y[sa[i]] && y[sa[i - 1] + k] == y[sa[i] +
22                 k] )
23                 x[sa[i]] = p;
24             else
25                 x[sa[i]] = ++p;
26             if ( ( m = p ) >= n )
27                 break;
28     }
29     REP( i, 1, n ) rk[sa[i]] = i;
30     for ( int j = 0, k = 0, i = 1; i <= n; ht[rk[i++]] = k )
31         for ( k ? k-- : 0, j = sa[rk[i] - 1]; str[i + k] == str[j + k];
32             k++ )

```

32 | }

5.7 AC 自动机

```
1 // 洛谷模版题的代码为例
2
3 const int N = 2e6 + 100, M = 26;
4
5 int mp( char ch ) { return ch - 'a'; }
6
7 struct ACA {
8     int ch[N][M], fail[N];
9     int ridx[N], in[N];
10    int sz;
11    vector<int> topo; // 把拓扑序预处理出来 以便如果多次查询
12    void init() {
13        sz = 1;
14        memset( ch[0], 0, sizeof ch[0] );
15        in[0] = fail[0] = 0;
16    }
17    void insert( const string& s, int id ) {
18        int n = s.size();
19        int u = 0, c;
20        for ( int i = 0; i < n; ++i ) {
21            c = mp( s[i] );
22            if ( !ch[u][c] ) {
23                memset( ch[sz], 0, sizeof ch[sz] );
24                in[sz] = fail[sz] = 0;
25                ch[u][c] = sz++;
26            }
27            u = ch[u][c];
28        }
29        ridx[id] = u;
30    }
31    void build() {
32        queue<int> Q;
33        fail[0] = 0;
34        for ( int c = 0, u; c < M; c++ ) {
35            u = ch[0][c];
36            if ( u ) {
```

```

37         Q.push( u );
38         fail[u] = 0;
39         in[0]++;
40     }
41 }
42 while ( !Q.empty() ) {
43     int r = Q.front();
44     Q.pop();
45     for ( int c = 0, u; c < M; c++ ) {
46         u = ch[r][c];
47         if ( !u ) {
48             ch[r][c] = ch[fail[r]][c];
49             continue;
50         }
51         fail[u] = ch[fail[r]][c];
52         in[fail[u]]++;
53         Q.push( u );
54     }
55 }
56 }
57 void calc_topo() {
58     queue<int> Q;
59     for ( int i = 0; i < sz; ++i )
60         if ( in[i] == 0 )
61             Q.push( i );
62     while ( !Q.empty() ) {
63         int u = Q.front();
64         Q.pop();
65         topo.push_back( u );
66         if ( !u ) continue;
67         in[fail[u]]--;
68         if ( in[fail[u]] == 0 )
69             Q.push( fail[u] );
70     }
71 }
72 } ac;
73
74 int n;
75 char s[N];
76 int cnt[N];

```

```

77
78 int main() {
79     n = rd();
80     ac.init();
81     for ( int i = 1; i <= n; ++i ) {
82         scanf( "%s", s );
83         ac.insert( s, i );
84     }
85     ac.build();
86     ac.calc_topo();
87     scanf( "%s", s );
88     int u = 0, len = strlen( s );
89     for ( int i = 0; i < len; ++i ) {
90         u = ac.ch[u][mp( s[i] )];
91         ++cnt[u];
92     }
93     for ( int i : ac.topo )
94         cnt[ac.fail[i]] += cnt[i];
95     for ( int i = 1; i <= n; ++i )
96         printf( "%d\n", cnt[ac.riidx[i]] );
97     return 0;
98 }

```

6 计算几何

6.1 计算几何模板 - 吉如一

我也看不懂先放上来再说

```

1 #define mp make_pair
2 #define fi first
3 #define se second
4 #define pb push_back
5 typedef double db;
6 const db eps = 1e-6;
7 const db pi = acos(-1);
8 int sign(db k) {
9     if (k > eps) return 1; else if (k < -eps) return -1; return 0;
10 }
11 int cmp(db k1, db k2) {return sign(k1 - k2);}

```

```

12 int inmid(db k1, db k2, db k3) {return sign(k1 - k3) * sign(k2 - k3
    ) <= 0;} // k3 在 [k1,k2] 内
13 struct point {
14     db x, y;
15     point operator + (const point &k1) const {return (point) {k1.x +
        x, k1.y + y};}
16     point operator - (const point &k1) const {return (point) {x - k1.
        x, y - k1.y};}
17     point operator * (db k1) const {return (point) {x*k1, y*k1};}
18     point operator / (db k1) const {return (point) {x / k1, y / k1};}
19     int operator == (const point &k1) const {return cmp(x, k1.x) == 0
        && cmp(y, k1.y) == 0;}
20     // 逆时针旋转
21     point turn(db k1) {return (point) {x*cos(k1) - y*sin(k1), x*sin(
        k1) + y*cos(k1)};}
22     point turn90() {return (point) { -y, x};}
23     bool operator < (const point k1) const {
24         int a = cmp(x, k1.x);
25         if (a == -1) return 1; else if (a == 1) return 0; else return
            cmp(y, k1.y) == -1;
26     }
27     db abs() {return sqrt(x * x + y * y);}
28     db abs2() {return x * x + y * y;}
29     db dis(point k1) {return ((*this) - k1).abs();}
30     point unit() {db w = abs(); return (point) {x / w, y / w};}
31     void scan() {double k1, k2; scanf("%lf%lf", &k1, &k2); x = k1; y
        = k2;}
32     void print() {printf("%.11lf %.11lf\n", x, y);}
33     db getw() {return atan2(y, x);}
34     point getdel() {if (sign(x) == -1 || (sign(x) == 0 && sign(y) ==
        -1)) return (*this) * (-1); else return (*this);}
35     int getP() const {return sign(y) == 1 || (sign(y) == 0 && sign(x)
        == -1);}
36 };
37 int inmid(point k1, point k2, point k3) {return inmid(k1.x, k2.x,
    k3.x) && inmid(k1.y, k2.y, k3.y);}
38 db cross(point k1, point k2) {return k1.x * k2.y - k1.y * k2.x;}
39 db dot(point k1, point k2) {return k1.x * k2.x + k1.y * k2.y;}
40 db rad(point k1, point k2) {return atan2(cross(k1, k2), dot(k1, k2)
    );}

```

```

41 // -pi -> pi
42 int compareangle (point k1, point k2) {
43     return k1.getP() < k2.getP() || (k1.getP() == k2.getP() && sign(
        cross(k1, k2)) > 0);
44 }
45 point proj(point k1, point k2, point q) { // q 到直线 k1,k2 的投影
46     point k = k2 - k1; return k1 + k * (dot(q - k1, k) / k.abs2());
47 }
48 point reflect(point k1, point k2, point q) {return proj(k1, k2, q)
    * 2 - q;}
49 int clockwise(point k1, point k2, point k3) { // k1 k2 k3 逆时针 1
    顺时针 -1 否则 0
50     return sign(cross(k2 - k1, k3 - k1));
51 }
52 int checkLL(point k1, point k2, point k3, point k4) { // 求直线 (L)
    线段 (S)k1,k2 和 k3,k4 的交点
53     return cmp(cross(k3 - k1, k4 - k1), cross(k3 - k2, k4 - k2)) !=
        0;
54 }
55 point getLL(point k1, point k2, point k3, point k4) {
56     db w1 = cross(k1 - k3, k4 - k3), w2 = cross(k4 - k3, k2 - k3);
        return (k1 * w2 + k2 * w1) / (w1 + w2);
57 }
58 int intersect(db l1, db r1, db l2, db r2) {
59     if (l1 > r1) swap(l1, r1); if (l2 > r2) swap(l2, r2); return cmp(
        r1, l2) != -1 && cmp(r2, l1) != -1;
60 }
61 int checkSS(point k1, point k2, point k3, point k4) {
62     return intersect(k1.x, k2.x, k3.x, k4.x) && intersect(k1.y, k2.y,
        k3.y, k4.y) &&
63         sign(cross(k3 - k1, k4 - k1)) * sign(cross(k3 - k2, k4 -
            k2)) <= 0 &&
64         sign(cross(k1 - k3, k2 - k3)) * sign(cross(k1 - k4, k2 -
            k4)) <= 0;
65 }
66 db disSP(point k1, point k2, point q) {
67     point k3 = proj(k1, k2, q);
68     if (inmid(k1, k2, k3)) return q.dis(k3); else return min(q.dis(k1
        ), q.dis(k2));
69 }

```

```

70 db disSS(point k1, point k2, point k3, point k4) {
71     if (checkSS(k1, k2, k3, k4)) return 0;
72     else return min(min(disSP(k1, k2, k3), disSP(k1, k2, k4)), min(
        disSP(k3, k4, k1), disSP(k3, k4, k2)));
73 }
74 int onS(point k1, point k2, point q) {return inmid(k1, k2, q) &&
    sign(cross(k1 - q, k2 - k1)) == 0;}
75 struct circle {
76     point o; db r;
77     void scan() {o.scan(); scanf("%lf", &r);}
78     int inside(point k) {return cmp(r, o.dis(k));}
79 };
80 struct line {
81     // p[0]->p[1]
82     point p[2];
83     line(point k1, point k2) {p[0] = k1; p[1] = k2;}
84     point& operator [] (int k) {return p[k];}
85     int include(point k) {return sign(cross(p[1] - p[0], k - p[0])) >
        0;}
86     point dir() {return p[1] - p[0];}
87     line push() { // 向外 ( 左手边 ) 平移 eps
88         const db eps = 1e-6;
89         point delta = (p[1] - p[0]).turn90().unit() * eps;
90         return {p[0] - delta, p[1] - delta};
91     }
92 };
93 point getLL(line k1, line k2) {return getLL(k1[0], k1[1], k2[0], k2
    [1]);}
94 int parallel(line k1, line k2) {return sign(cross(k1.dir(), k2.dir
    ())) == 0;}
95 int sameDir(line k1, line k2) {return parallel(k1, k2) && sign(dot(
    k1.dir(), k2.dir())) == 1;}
96 int operator < (line k1, line k2) {
97     if (sameDir(k1, k2)) return k2.include(k1[0]);
98     return compareangle(k1.dir(), k2.dir());
99 }
100 int checkpos(line k1, line k2, line k3) {return k3.include(getLL(k1
    , k2));}
101 vector<line> getHL(vector<line> &L) { // 求半平面交 , 半平面是逆时
    针方向 , 输出按照逆时针

```

```

102 sort(L.begin(), L.end()); deque<line> q;
103 for (int i = 0; i < (int)L.size(); i++) {
104     if (i && sameDir(L[i], L[i - 1])) continue;
105     while (q.size() > 1 && !checkpos(q[q.size() - 2], q[q.size() -
106         1], L[i])) q.pop_back();
107     while (q.size() > 1 && !checkpos(q[1], q[0], L[i])) q.pop_front
108         ();
109     q.push_back(L[i]);
110 }
111 while (q.size() > 2 && !checkpos(q[q.size() - 2], q[q.size() -
112     1], q[0])) q.pop_back();
113 while (q.size() > 2 && !checkpos(q[1], q[0], q[q.size() - 1])) q.
114     pop_front();
115 vector<line>ans; for (int i = 0; i < q.size(); i++) ans.push_back
116     (q[i]);
117 return ans;
118 }
119 db closepoint(vector<point>&A, int l, int r) { // 最近点对 , 先要按
120     照 x 坐标排序
121     if (r - l <= 5) {
122         db ans = 1e20;
123         for (int i = l; i <= r; i++) for (int j = i + 1; j <= r; j++)
124             ans = min(ans, A[i].dis(A[j]));
125         return ans;
126     }
127     int mid = l + r >> 1; db ans = min(closepoint(A, l, mid),
128         closepoint(A, mid + 1, r));
129     vector<point>B; for (int i = l; i <= r; i++) if (abs(A[i].x - A[
130         mid].x) <= ans) B.push_back(A[i]);
131     sort(B.begin(), B.end(), [](point k1, point k2) {return k1.y < k2
132         .y;});
133     for (int i = 0; i < B.size(); i++) for (int j = i + 1; j < B.size
134         () && B[j].y - B[i].y < ans; j++) ans = min(ans, B[i].dis(B[j
135         ]));
136     return ans;
137 }
138 int checkposCC(circle k1, circle k2) { // 返回两个圆的公切线数量
139     if (cmp(k1.r, k2.r) == -1) swap(k1, k2);
140     db dis = k1.o.dis(k2.o); int w1 = cmp(dis, k1.r + k2.r), w2 =
141         cmp(dis, k1.r - k2.r);

```

```

129     if (w1 > 0) return 4; else if (w1 == 0) return 3; else if (w2 >
        0) return 2;
130     else if (w2 == 0) return 1; else return 0;
131 }
132 vector<point> getCL(circle k1, point k2, point k3) { // 沿着 k2->k3
        方向给出 , 相切给出两个
133     point k = proj(k2, k3, k1.o); db d = k1.r * k1.r - (k - k1.o).
        abs2();
134     if (sign(d) == -1) return {};
135     point del = (k3 - k2).unit() * sqrt(max((db)0.0, d)); return {k -
        del, k + del};
136 }
137 vector<point> getCC(circle k1, circle k2) { // 沿圆 k1 逆时针给出 ,
        相切给出两个
138     int pd = checkposCC(k1, k2); if (pd == 0 || pd == 4) return {};
139     db a = (k2.o - k1.o).abs2(), cosA = (k1.r * k1.r + a - k2.r * k2.
        r) / (2 * k1.r * sqrt(max(a, (db)0.0)));
140     db b = k1.r * cosA, c = sqrt(max((db)0.0, k1.r * k1.r - b * b));
141     point k = (k2.o - k1.o).unit(), m = k1.o + k * b, del = k.turn90
        () * c;
142     return {m - del, m + del};
143 }
144 vector<point> TangentCP(circle k1, point k2) { // 沿圆 k1 逆时针给
        出
145     db a = (k2 - k1.o).abs(), b = k1.r * k1.r / a, c = sqrt(max((db)
        0.0, k1.r * k1.r - b * b));
146     point k = (k2 - k1.o).unit(), m = k1.o + k * b, del = k.turn90()
        * c;
147     return {m - del, m + del};
148 }
149 vector<line> TangentoutCC(circle k1, circle k2) {
150     int pd = checkposCC(k1, k2); if (pd == 0) return {};
151     if (pd == 1) {point k = getCC(k1, k2)[0]; return {(line) {k, k
        }};}
152     if (cmp(k1.r, k2.r) == 0) {
153         point del = (k2.o - k1.o).unit().turn90().getdel();
154         return {(line){k1.o - del * k1.r, k2.o - del * k2.r}, (line) {
            k1.o + del*k1.r, k2.o + del*k2.r}};
155     } else {
156         point p = (k2.o * k1.r - k1.o * k2.r) / (k1.r - k2.r);

```

```

157     vector<point>A = TangentCP(k1, p), B = TangentCP(k2, p);
158     vector<line>ans; for (int i = 0; i < A.size(); i++) ans.
        push_back((line) {A[i], B[i]});
159     return ans;
160 }
161 }
162 vector<line> TangentinCC(circle k1, circle k2) {
163     int pd = checkposCC(k1, k2); if (pd <= 2) return {};
164     if (pd == 3) {point k = getCC(k1, k2)[0]; return {(line) {k, k
        }};};
165     point p = (k2.o * k1.r + k1.o * k2.r) / (k1.r + k2.r);
166     vector<point>A = TangentCP(k1, p), B = TangentCP(k2, p);
167     vector<line>ans; for (int i = 0; i < A.size(); i++) ans.push_back
        ((line) {A[i], B[i]});
168     return ans;
169 }
170 vector<line> TangentCC(circle k1, circle k2) {
171     int flag = 0; if (k1.r < k2.r) swap(k1, k2), flag = 1;
172     vector<line>A = TangentoutCC(k1, k2), B = TangentinCC(k1, k2);
173     for (line k : B) A.push_back(k);
174     if (flag) for (line &k : A) swap(k[0], k[1]);
175     return A;
176 }
177 db getarea(circle k1, point k2, point k3) {
178     // 圓 k1 與三角形 k2 k3 k1.o 的有向面積交
179     point k = k1.o; k1.o = k1.o - k; k2 = k2 - k; k3 = k3 - k;
180     int pd1 = k1.inside(k2), pd2 = k1.inside(k3);
181     vector<point>A = getCL(k1, k2, k3);
182     if (pd1 >= 0) {
183         if (pd2 >= 0) return cross(k2, k3) / 2;
184         return k1.r * k1.r * rad(A[1], k3) / 2 + cross(k2, A[1]) / 2;
185     } else if (pd2 >= 0) {
186         return k1.r * k1.r * rad(k2, A[0]) / 2 + cross(A[0], k3) / 2;
187     } else {
188         int pd = cmp(k1.r, disSP(k2, k3, k1.o));
189         if (pd <= 0) return k1.r * k1.r * rad(k2, k3) / 2;
190         return cross(A[0], A[1]) / 2 + k1.r * k1.r * (rad(k2, A[0]) +
            rad(A[1], k3)) / 2;
191     }
192 }

```

```

193 circle getcircle(point k1, point k2, point k3) {
194     db a1 = k2.x - k1.x, b1 = k2.y - k1.y, c1 = (a1 * a1 + b1 * b1) /
        2;
195     db a2 = k3.x - k1.x, b2 = k3.y - k1.y, c2 = (a2 * a2 + b2 * b2) /
        2;
196     db d = a1 * b2 - a2 * b1;
197     point o = (point) {k1.x + (c1 * b2 - c2 * b1) / d, k1.y + (a1 *
        c2 - a2 * c1) / d};
198     return (circle) {o, k1.dis(o)};
199 }
200 circle getScircle(vector<point> A) {
201     random_shuffle(A.begin(), A.end());
202     circle ans = (circle) {A[0], 0};
203     for (int i = 1; i < A.size(); i++)
204         if (ans.inside(A[i]) == -1) {
205             ans = (circle) {A[i], 0};
206             for (int j = 0; j < i; j++)
207                 if (ans.inside(A[j]) == -1) {
208                     ans.o = (A[i] + A[j]) / 2; ans.r = ans.o.dis(A[i]);
209                     for (int k = 0; k < j; k++)
210                         if (ans.inside(A[k]) == -1)
211                             ans = getcircle(A[i], A[j], A[k]);
212                 }
213         }
214     return ans;
215 }
216 db area(vector<point> A) { // 多边形用 vector<point> 表示 , 逆时针
217     db ans = 0;
218     for (int i = 0; i < A.size(); i++) ans += cross(A[i], A[(i + 1) %
        A.size()]);
219     return ans / 2;
220 }
221 int checkconvex(vector<point>A) {
222     int n = A.size(); A.push_back(A[0]); A.push_back(A[1]);
223     for (int i = 0; i < n; i++) if (sign(cross(A[i + 1] - A[i], A[i +
        2] - A[i])) == -1) return 0;
224     return 1;
225 }
226 int contain(vector<point>A, point q) { // 2 内部 1 边界 0 外部
227     int pd = 0; A.push_back(A[0]);

```

```

228     for (int i = 1; i < A.size(); i++) {
229         point u = A[i - 1], v = A[i];
230         if (onS(u, v, q)) return 1; if (cmp(u.y, v.y) > 0) swap(u, v);
231         if (cmp(u.y, q.y) >= 0 || cmp(v.y, q.y) < 0) continue;
232         if (sign(cross(u - v, q - v)) < 0) pd ^= 1;
233     }
234     return pd << 1;
235 }
236 vector<point> ConvexHull(vector<point>A, int flag = 1) { // flag=0
    不严格 flag=1 严格
237     int n = A.size(); vector<point>ans(n * 2);
238     sort(A.begin(), A.end()); int now = -1;
239     for (int i = 0; i < A.size(); i++) {
240         while (now > 0 && sign(cross(ans[now] - ans[now - 1], A[i] -
            ans[now - 1])) < flag) now--;
241         ans[++now] = A[i];
242     } int pre = now;
243     for (int i = n - 2; i >= 0; i--) {
244         while (now > pre && sign(cross(ans[now] - ans[now - 1], A[i] -
            ans[now - 1])) < flag) now--;
245         ans[++now] = A[i];
246     } ans.resize(now); return ans;
247 }
248 db convexDiameter(vector<point>A) {
249     int now = 0, n = A.size(); db ans = 0;
250     for (int i = 0; i < A.size(); i++) {
251         now = max(now, i);
252         while (1) {
253             db k1 = A[i].dis(A[now % n]), k2 = A[i].dis(A[(now + 1) % n])
                ;
254             ans = max(ans, max(k1, k2)); if (k2 > k1) now++; else break;
255         }
256     }
257     return ans;
258 }
259 vector<point> convexcut(vector<point>A, point k1, point k2) {
260     // 保留 k1,k2,p 逆时针的所有点
261     int n = A.size(); A.push_back(A[0]); vector<point>ans;
262     for (int i = 0; i < n; i++) {
263         int w1 = clockwise(k1, k2, A[i]), w2 = clockwise(k1, k2, A[i +

```

```

1]);
264     if (w1 >= 0) ans.push_back(A[i]);
265     if (w1 * w2 < 0) ans.push_back(getLL(k1, k2, A[i], A[i + 1]));
266 }
267 return ans;
268 }
269 int checkPoS(vector<point>A, point k1, point k2) {
270     // 多边形 A 和直线 ( 线段 )k1->k2 严格相交 , 注释部分为线段
271     struct ins {
272         point m, u, v;
273         int operator < (const ins& k) const {return m < k.m;}
274     }; vector<ins>B;
275     //if (contain(A,k1)==2||contain(A,k2)==2) return 1;
276     vector<point>poly = A; A.push_back(A[0]);
277     for (int i = 1; i < A.size(); i++) if (checkLL(A[i - 1], A[i], k1
278         , k2)) {
279         point m = getLL(A[i - 1], A[i], k1, k2);
280         if (inmid(A[i - 1], A[i], m)/ *&&inmid(k1,k2,m)*/) B.push_back
281             ((ins) {m, A[i - 1], A[i]});
282     }
283     if (B.size() == 0) return 0; sort(B.begin(), B.end());
284     int now = 1; while (now < B.size() && B[now].m == B[0].m) now++;
285     if (now == B.size()) return 0;
286     int flag = contain(poly, (B[0].m + B[now].m) / 2);
287     if (flag == 2) return 1;
288     point d = B[now].m - B[0].m;
289     for (int i = now; i < B.size(); i++) {
290         if (!(B[i].m == B[i - 1].m) && flag == 2) return 1;
291         int tag = sign(cross(B[i].v - B[i].u, B[i].m + d - B[i].u));
292         if (B[i].m == B[i].u || B[i].m == B[i].v) flag += tag; else
293             flag += tag * 2;
294     }
295     //return 0;
296     return flag == 2;
297 }
298 int checkinp(point r, point l, point m) {
299     if (compareangle(l, r)) {return compareangle(l, m) &&
300         compareangle(m, r);}
301     return compareangle(l, m) || compareangle(m, r);
302 }

```

```

299 int checkPosFast(vector<point>A, point k1, point k2) { // 快速检查
    线段是否和多边形严格相交
300 if (contain(A, k1) == 2 || contain(A, k2) == 2) return 1; if (k1
    == k2) return 0;
301 A.push_back(A[0]); A.push_back(A[1]);
302 for (int i = 1; i + 1 < A.size(); i++)
303     if (checkLL(A[i - 1], A[i], k1, k2)) {
304         point now = getLL(A[i - 1], A[i], k1, k2);
305         if (inmid(A[i - 1], A[i], now) == 0 || inmid(k1, k2, now) ==
            0) continue;
306         if (now == A[i]) {
307             if (A[i] == k2) continue;
308             point pre = A[i - 1], ne = A[i + 1];
309             if (checkinp(pre - now, ne - now, k2 - now)) return 1;
310         } else if (now == k1) {
311             if (k1 == A[i - 1] || k1 == A[i]) continue;
312             if (checkinp(A[i - 1] - k1, A[i] - k1, k2 - k1)) return 1;
313         } else if (now == k2 || now == A[i - 1]) continue;
314         else return 1;
315     }
316 return 0;
317 }
318 // 拆分凸包成上下凸壳 凸包尽量都随机旋转一个角度来避免出现相同横坐
    标
319 // 尽量特判只有一个点的情况 凸包逆时针
320 void getUDP(vector<point>A, vector<point>&U, vector<point>&D) {
321     db l = 1e100, r = -1e100;
322     for (int i = 0; i < A.size(); i++) l = min(l, A[i].x), r = max(r,
        A[i].x);
323     int wherel, wherer;
324     for (int i = 0; i < A.size(); i++) if (cmp(A[i].x, l) == 0)
        wherel = i;
325     for (int i = A.size(); i; i--) if (cmp(A[i - 1].x, r) == 0)
        wherer = i - 1;
326     U.clear(); D.clear(); int now = wherel;
327     while (1) {D.push_back(A[now]); if (now == wherer) break; now++;
        if (now >= A.size()) now = 0;}
328     now = wherel;
329     while (1) {U.push_back(A[now]); if (now == wherer) break; now--;
        if (now < 0) now = A.size() - 1;}

```

```

330 }
331 // 需要保证凸包点数大于等于 3, 2 内部, 1 边界, 0 外部
332 int containCoP(const vector<point>&U, const vector<point>&D, point
    k) {
333     db lx = U[0].x, rx = U[U.size() - 1].x;
334     if (k == U[0] || k == U[U.size() - 1]) return 1;
335     if (cmp(k.x, lx) == -1 || cmp(k.x, rx) == 1) return 0;
336     int where1 = lower_bound(U.begin(), U.end(), (point) {k.x, -1e100
        }) - U.begin();
337     int where2 = lower_bound(D.begin(), D.end(), (point) {k.x, -1e100
        }) - D.begin();
338     int w1 = clockwise(U[where1 - 1], U[where1], k), w2 = clockwise(D
        [where2 - 1], D[where2], k);
339     if (w1 == 1 || w2 == -1) return 0; else if (w1 == 0 || w2 == 0)
        return 1; return 2;
340 }
341 // d 是方向, 输出上方切点和下方切点
342 pair<point, point> getTangentCow(const vector<point> &U, const
    vector<point> &D, point d) {
343     if (sign(d.x) < 0 || (sign(d.x) == 0 && sign(d.y) < 0)) d = d *
        (-1);
344     point whereU, whereD;
345     if (sign(d.x) == 0) return mp(U[0], U[U.size() - 1]);
346     int l = 0, r = U.size() - 1, ans = 0;
347     while (l < r) {int mid = l + r >> 1; if (sign(cross(U[mid + 1] -
        U[mid], d)) <= 0) l = mid + 1, ans = mid + 1; else r = mid;}
348     whereU = U[ans]; l = 0, r = D.size() - 1, ans = 0;
349     while (l < r) {int mid = l + r >> 1; if (sign(cross(D[mid + 1] -
        D[mid], d)) >= 0) l = mid + 1, ans = mid + 1; else r = mid;}
350     whereD = D[ans]; return mp(whereU, whereD);
351 }
352 // 先检查 contain, 逆时针给出
353 pair<point, point> getTangentCoP(const vector<point>&U, const
    vector<point>&D, point k) {
354     db lx = U[0].x, rx = U[U.size() - 1].x;
355     if (k.x < lx) {
356         int l = 0, r = U.size() - 1, ans = U.size() - 1;
357         while (l < r) {int mid = l + r >> 1; if (clockwise(k, U[mid], U
            [mid + 1]) == 1) l = mid + 1; else ans = mid, r = mid;}
358         point w1 = U[ans]; l = 0, r = D.size() - 1, ans = D.size() - 1;

```

```

359     while (l < r) {int mid = l + r >> 1; if (clockwise(k, D[mid], D
        [mid + 1]) == -1) l = mid + 1; else ans = mid, r = mid;}
360     point w2 = D[ans]; return mp(w1, w2);
361 } else if (k.x > rx) {
362     int l = 1, r = U.size(), ans = 0;
363     while (l < r) {int mid = l + r >> 1; if (clockwise(k, U[mid], U
        [mid - 1]) == -1) r = mid; else ans = mid, l = mid + 1;}
364     point w1 = U[ans]; l = 1, r = D.size(), ans = 0;
365     while (l < r) {int mid = l + r >> 1; if (clockwise(k, D[mid], D
        [mid - 1]) == 1) r = mid; else ans = mid, l = mid + 1;}
366     point w2 = D[ans]; return mp(w2, w1);
367 } else {
368     int where1 = lower_bound(U.begin(), U.end(), (point) {k.x, -1
        e100}) - U.begin();
369     int where2 = lower_bound(D.begin(), D.end(), (point) {k.x, -1
        e100}) - D.begin();
370     if ((k.x == lx && k.y > U[0].y) || (where1 && clockwise(U[
        where1 - 1], U[where1], k) == 1)) {
371         int l = 1, r = where1 + 1, ans = 0;
372         while (l < r) {int mid = l + r >> 1; if (clockwise(k, U[mid],
            U[mid - 1]) == 1) ans = mid, l = mid + 1; else r = mid;}
373         point w1 = U[ans]; l = where1, r = U.size() - 1, ans = U.size
            () - 1;
374         while (l < r) {int mid = l + r >> 1; if (clockwise(k, U[mid],
            U[mid + 1]) == 1) l = mid + 1; else ans = mid, r = mid;}
375         point w2 = U[ans]; return mp(w2, w1);
376     } else {
377         int l = 1, r = where2 + 1, ans = 0;
378         while (l < r) {int mid = l + r >> 1; if (clockwise(k, D[mid],
            D[mid - 1]) == -1) ans = mid, l = mid + 1; else r = mid;}
379         point w1 = D[ans]; l = where2, r = D.size() - 1, ans = D.size
            () - 1;
380         while (l < r) {int mid = l + r >> 1; if (clockwise(k, D[mid],
            D[mid + 1]) == -1) l = mid + 1; else ans = mid, r = mid;}
381         point w2 = D[ans]; return mp(w1, w2);
382     }
383 }
384 }
385 struct P3 {
386     db x, y, z;

```

```

387 P3 operator + (P3 k1) {return (P3) {x + k1.x, y + k1.y, z + k1.z
    };}
388 P3 operator - (P3 k1) {return (P3) {x - k1.x, y - k1.y, z - k1.z
    };}
389 P3 operator * (db k1) {return (P3) {x*k1, y*k1, z*k1};}
390 P3 operator / (db k1) {return (P3) {x / k1, y / k1, z / k1};}
391 db abs2() {return x * x + y * y + z * z;}
392 db abs() {return sqrt(x * x + y * y + z * z);}
393 P3 unit() {return (*this) / abs();}
394 int operator < (const P3 k1) const {
395     if (cmp(x, k1.x) != 0) return x < k1.x;
396     if (cmp(y, k1.y) != 0) return y < k1.y;
397     return cmp(z, k1.z) == -1;
398 }
399 int operator == (const P3 k1) {
400     return cmp(x, k1.x) == 0 && cmp(y, k1.y) == 0 && cmp(z, k1.z)
        == 0;
401 }
402 void scan() {
403     double k1, k2, k3; scanf("%lf%lf%lf", &k1, &k2, &k3);
404     x = k1; y = k2; z = k3;
405 }
406 };
407 P3 cross(P3 k1, P3 k2) {return (P3) {k1.y*k2.z - k1.z*k2.y, k1.z*k2
    .x - k1.x*k2.z, k1.x*k2.y - k1.y*k2.x};}
408 db dot(P3 k1, P3 k2) {return k1.x * k2.x + k1.y * k2.y + k1.z * k2.
    z;}
409 //p=(3,4,5),l=(13,19,21),theta=85 ans=(2.83,4.62,1.77)
410 P3 turn3D(db k1, P3 l, P3 p) {
411     l = l.unit(); P3 ans; db c = cos(k1), s = sin(k1);
412     ans.x = p.x * (l.x * l.x * (1 - c) + c) + p.y * (l.x * l.y * (1 -
        c) - l.z * s) + p.z * (l.x * l.z * (1 - c) + l.y * s);
413     ans.y = p.x * (l.x * l.y * (1 - c) + l.z * s) + p.y * (l.y * l.y
        * (1 - c) + c) + p.z * (l.y * l.z * (1 - c) - l.x * s);
414     ans.z = p.x * (l.x * l.z * (1 - c) - l.y * s) + p.y * (l.y * l.z
        * (1 - c) + l.x * s) + p.z * (l.x * l.x * (1 - c) + c);
415     return ans;
416 }
417 typedef vector<P3> VP;
418 typedef vector<VP> VVP;

```

```

419 db Acos(db x) {return acos(max(-(db)1, min(x, (db)1)));}
420 // 球面距离 , 圆心原点 , 半径 1
421 db Odist(P3 a, P3 b) {db r = Acos(dot(a, b)); return r;}
422 db r; P3 rnd;
423 vector<db> solve(db a, db b, db c) {
424     db r = sqrt(a * a + b * b), th = atan2(b, a);
425     if (cmp(c, -r) == -1) return {0};
426     else if (cmp(r, c) <= 0) return {1};
427     else {
428         db tr = pi - Acos(c / r); return {th + pi - tr, th + pi + tr};
429     }
430 }
431 vector<db> jiao(P3 a, P3 b) {
432     // dot(rd+x*cos(t)+y*sin(t),b) >= cos(r)
433     if (cmp(Odist(a, b), 2 * r) > 0) return {0};
434     P3 rd = a * cos(r), z = a.unit(), y = cross(z, rnd).unit(), x =
        cross(y, z).unit();
435     vector<db> ret = solve(-(dot(x, b) * sin(r)), -(dot(y, b) * sin(r)
        )), -(cos(r) - dot(rd, b)));
436     return ret;
437 }
438 db norm(db x, db l = 0, db r = 2 * pi) { // change x into [l,r)
439     while (cmp(x, l) == -1) x += (r - l); while (cmp(x, r) >= 0) x -=
        (r - l);
440     return x;
441 }
442 db disLP(P3 k1, P3 k2, P3 q) {
443     return (cross(k2 - k1, q - k1)).abs() / (k2 - k1).abs();
444 }
445 db disLL(P3 k1, P3 k2, P3 k3, P3 k4) {
446     P3 dir = cross(k2 - k1, k4 - k3); if (sign(dir.abs()) == 0)
        return disLP(k1, k2, k3);
447     return fabs(dot(dir.unit(), k1 - k2));
448 }
449 VP getFL(P3 p, P3 dir, P3 k1, P3 k2) {
450     db a = dot(k2 - p, dir), b = dot(k1 - p, dir), d = a - b;
451     if (sign(fabs(d)) == 0) return {};
452     return {(k1 * a - k2 * b) / d};
453 }
454 VP getFF(P3 p1, P3 dir1, P3 p2, P3 dir2) { // 返回一条线

```

```

455     P3 e = cross(dir1, dir2), v = cross(dir1, e);
456     db d = dot(dir2, v); if (sign(abs(d)) == 0) return {};
457     P3 q = p1 + v * dot(dir2, p2 - p1) / d; return {q, q + e};
458 }
459 // 3D Covex Hull Template
460 db getV(P3 k1, P3 k2, P3 k3, P3 k4) { // get the Volume
461     return dot(cross(k2 - k1, k3 - k1), k4 - k1);
462 }
463 db rand_db() {return 1.0 * rand() / RAND_MAX;}
464 VP convexHull2D(VP A, P3 dir) {
465     P3 x = {(db)rand(), (db)rand(), (db)rand()}; x = x.unit();
466     x = cross(x, dir).unit(); P3 y = cross(x, dir).unit();
467     P3 vec = dir.unit() * dot(A[0], dir);
468     vector<point>B;
469     for (int i = 0; i < A.size(); i++) B.push_back((point) {dot(A[i],
470         x), dot(A[i], y)});
471     B = ConvexHull(B); A.clear();
472     for (int i = 0; i < B.size(); i++) A.push_back(x * B[i].x + y * B
473         [i].y + vec);
474     return A;
475 }
476 namespace CH3 {
477 VVP ret; set<pair<int, int> >e;
478 int n; VP p, q;
479 void wrap(int a, int b) {
480     if (e.find({a, b}) == e.end()) {
481         int c = -1;
482         for (int i = 0; i < n; i++) if (i != a && i != b) {
483             if (c == -1 || sign(getV(q[c], q[a], q[b], q[i])) > 0) c =
484                 i;
485         }
486         if (c != -1) {
487             ret.push_back({p[a], p[b], p[c]});
488             e.insert({a, b}); e.insert({b, c}); e.insert({c, a});
489             wrap(c, b); wrap(a, c);
490         }
491     }
492 }
493 VVP ConvexHull3D(VP _p) {
494     p = q = _p; n = p.size();

```

```

492   ret.clear(); e.clear();
493   for (auto &i : q) i = i + (P3) {rand_db() * 1e-4, rand_db() * 1e
      -4, rand_db() * 1e-4};
494   for (int i = 1; i < n; i++) if (q[i].x < q[0].x) swap(p[0], p[i])
      , swap(q[0], q[i]);
495   for (int i = 2; i < n; i++) if ((q[i].x - q[0].x) * (q[1].y - q
      [0].y) > (q[i].y - q[0].y) * (q[1].x - q[0].x)) swap(q[1], q[i]
      ), swap(p[1], p[i]);
496   wrap(0, 1);
497   return ret;
498 }
499 }
500 VVP reduceCH(VVP A) {
501   VVP ret; map<P3, VP> M;
502   for (VP nowF : A) {
503     P3 dir = cross(nowF[1] - nowF[0], nowF[2] - nowF[0]).unit();
504     for (P3 k1 : nowF) M[dir].pb(k1);
505   }
506   for (pair<P3, VP> nowF : M) ret.pb(convexHull2D(nowF.se, nowF.fi)
      );
507   return ret;
508 }
509 // 把一个面变成 ( 点 , 法向量 ) 的形式
510 pair<P3, P3> getF(VP F) {
511   return mp(F[0], cross(F[1] - F[0], F[2] - F[0]).unit());
512 }
513 // 3D Cut 保留 dot(dir,x-p)>=0 的部分
514 VVP ConvexCut3D(VVP A, P3 p, P3 dir) {
515   VVP ret; VP sec;
516   for (VP nowF : A) {
517     int n = nowF.size(); VP ans; int dif = 0;
518     for (int i = 0; i < n; i++) {
519       int d1 = sign(dot(dir, nowF[i] - p));
520       int d2 = sign(dot(dir, nowF[(i + 1) % n] - p));
521       if (d1 >= 0) ans.pb(nowF[i]);
522       if (d1 * d2 < 0) {
523         P3 q = getFL(p, dir, nowF[i], nowF[(i + 1) % n])[0];
524         ans.push_back(q); sec.push_back(q);
525       }
526       if (d1 == 0) sec.push_back(nowF[i]); else dif = 1;

```

```

527         dif |= (sign(dot(dir, cross(nowF[(i + 1) % n] - nowF[i], nowF
           [(i + 1) % n] - nowF[i]))) == -1);
528     }
529     if (ans.size() > 0 && dif) ret.push_back(ans);
530 }
531 if (sec.size() > 0) ret.push_back(convexHull2D(sec, dir));
532 return ret;
533 }
534 db vol(VVP A) {
535     if (A.size() == 0) return 0; P3 p = A[0][0]; db ans = 0;
536     for (VP nowF : A)
537         for (int i = 2; i < nowF.size(); i++)
538             ans += abs(getV(p, nowF[0], nowF[i - 1], nowF[i]));
539     return ans / 6;
540 }
541 VVP init(db INF) {
542     VVP pss(6, VP(4));
543     pss[0][0] = pss[1][0] = pss[2][0] = { -INF, -INF, -INF};
544     pss[0][3] = pss[1][1] = pss[5][2] = { -INF, -INF, INF};
545     pss[0][1] = pss[2][3] = pss[4][2] = { -INF, INF, -INF};
546     pss[0][2] = pss[5][3] = pss[4][1] = { -INF, INF, INF};
547     pss[1][3] = pss[2][1] = pss[3][2] = {INF, -INF, -INF};
548     pss[1][2] = pss[5][1] = pss[3][3] = {INF, -INF, INF};
549     pss[2][2] = pss[4][3] = pss[3][1] = {INF, INF, -INF};
550     pss[5][0] = pss[4][0] = pss[3][0] = {INF, INF, INF};
551     return pss;
552 }

```

7 STL

7.1 算法库

不修改序列的操作

批量操作

在标头 `<algorithm>` 定义

`for_each`

应用一元函数对象到范围中元素 (函数模板)

`ranges::for_each` (C++20)

应用一元函数对象到范围中元素 (算法函数对象)

`for_each_n` (C++17)

应用函数对象到序列的前 N 个元素 (函数模板)

`ranges::for_each_n` (C++20)

应用函数对象到序列的前 N 个元素 (算法函数对象)

搜索操作

在标头 `<algorithm>` 定义

`all_of` (C++11)

`any_of` (C++11)

`none_of` (C++11)

检查谓词是否对范围中所有、任一或无元素为 `true` (函数模板)

`ranges::all_of` (C++20)

`ranges::any_of` (C++20)

`ranges::none_of` (C++20)

检查谓词是否对范围中所有、任一或无元素为 `true` (算法函数对象)

`ranges::contains` (C++23)

`ranges::contains_subrange` (C++23)

检查范围是否包含给定元素或子范围 (算法函数对象)

`find`

`find_if`

`find_if_not` (C++11)

查找首个满足特定条件的元素 (函数模板)

`ranges::find` (C++20)

`ranges::find_if` (C++20)

`ranges::find_if_not` (C++20)

查找首个满足特定条件的元素 (算法函数对象)

`ranges::find_last` (C++23)

`ranges::find_last_if` (C++23)

`ranges::find_last_if_not` (C++23)

查找最后一个满足特定条件的元素 (算法函数对象)

`find_end`

查找元素序列在特定范围中最后一次出现 (函数模板)

`ranges::find_end` (C++20)

查找元素序列在特定范围中最后一次出现 (算法函数对象)

`find_first_of`

搜索一组元素中任一元素 (函数模板)

`ranges::find_first_of` (C++20)

搜索一组元素中任一元素 (算法函数对象)

`adjacent_find`

查找首对相同 (或满足给定谓词) 的相邻元素 (函数模板)

`ranges::adjacent_find` (C++20)

查找首对相同 (或满足给定谓词) 的相邻元素 (算法函数对象)

count
count_if
 返回满足特定条件的元素数目 (函数模板)
ranges::count (C++20)
ranges::count_if (C++20)
 返回满足特定条件的元素数目 (算法函数对象)
mismatch
 查找两个范围的首个不同之处 (函数模板)
ranges::mismatch (C++20)
 查找两个范围的首个不同之处 (算法函数对象)
equal
 判断两组元素是否相同 (函数模板)
ranges::equal (C++20)
 判断两组元素是否相同 (算法函数对象)
search
 搜索元素范围的首次出现 (函数模板)
ranges::search (C++20)
 搜索元素范围的首次出现 (算法函数对象)
search_n
 搜索元素在范围中首次连续若干次出现 (函数模板)
ranges::search_n (C++20)
 搜索元素在范围中首次连续若干次出现 (算法函数对象)
ranges::starts_with (C++23)
 检查一个范围是否始于另一范围 (算法函数对象)
ranges::ends_with (C++23)
 检查一个范围是否终于另一范围 (算法函数对象)
折叠操作 (C++23 起)
 在标头 `<algorithm>` 定义
ranges::fold_left (C++23)
 左折叠范围中元素 (算法函数对象)
ranges::fold_left_first (C++23)
 以首元素为初值左折叠范围中元素 (算法函数对象)
ranges::fold_right (C++23)
 右折叠范围中元素 (算法函数对象)
ranges::fold_right_last (C++23)
 以末元素为初值右折叠范围中元素 (算法函数对象)
ranges::fold_left_with_iter (C++23)
 左折叠范围中元素, 并返回 **pair** (迭代器, 值) (算法函数对象)
ranges::fold_left_first_with_iter (C++23)
 以首元素为初值左折叠范围中元素, 并返回 **pair** (迭代器, optional) (算法函数对象)

修改序列的操作

复制操作

在标头 `<algorithm>` 定义

`copy`

`copy_if` (C++11)

复制范围中元素到新位置 (函数模板)

`ranges::copy` (C++20)

`ranges::copy_if` (C++20)

复制范围中元素到新位置 (算法函数对象)

`copy_n` (C++11)

复制若干元素到新位置 (函数模板)

`ranges::copy_n` (C++20)

复制若干元素到新位置 (算法函数对象)

`copy_backward`

从后往前复制范围中元素 (函数模板)

`ranges::copy_backward` (C++20)

从后往前复制范围中元素 (算法函数对象)

`move` (C++11)

将范围中元素移到新位置 (函数模板)

`ranges::move` (C++20)

将范围中元素移到新位置 (算法函数对象)

`move_backward` (C++11)

从后往前将范围中元素移到新位置 (函数模板)

`ranges::move_backward` (C++20)

从后往前将范围中元素移到新位置 (算法函数对象)

交换操作

在标头 `<algorithm>` 定义 (C++11 前)

在标头 `<utility>` 定义 (C++11 起)

在标头 `<string_view>` 定义

`swap`

在标头 `<algorithm>` 定义

交换两个对象的值 (函数模板)

`swap_ranges`

交换两个范围的元素 (函数模板)

`ranges::swap_ranges` (C++20)

交换两个范围的元素 (算法函数对象)

`iter_swap`

交换两个迭代器所指向的元素 (函数模板)

变换操作

在标头 `<algorithm>` 定义

transform

应用函数到元素范围，并在目标范围存储结果 (函数模板)

ranges::transform (C++20)

应用函数到元素范围 (算法函数对象)

replace

replace_if

替换所有满足特定条件的值为另一个值 (函数模板)

ranges::replace (C++20)

ranges::replace_if (C++20)

替换所有满足特定条件的值为另一个值 (算法函数对象)

replace_copy

replace_copy_if

复制范围，并将满足特定条件的元素替换为另一个值 (函数模板)

ranges::replace_copy (C++20)

ranges::replace_copy_if (C++20)

复制范围，并将满足特定条件的元素替换为另一个值 (算法函数对象)

生成操作

在标头 `<algorithm>` 定义

fill

以复制的方式赋给定值到范围中所有元素 (函数模板)

ranges::fill (C++20)

赋给定值到范围中元素 (算法函数对象)

fill_n

以复制的方式赋给定值到范围中 N 个元素 (函数模板)

ranges::fill_n (C++20)

赋给定值到若干元素 (算法函数对象)

generate

赋连续函数调用结果到范围中所有元素 (函数模板)

ranges::generate (C++20)

将函数结果保存到范围中 (算法函数对象)

generate_n

赋连续函数调用结果到范围中 N 个元素 (函数模板)

ranges::generate_n (C++20)

保存 N 次函数应用的结果 (算法函数对象)

移除操作

在标头 `<algorithm>` 定义

remove

remove_if

移除满足特定条件的元素 (函数模板)

ranges::remove (C++20)

ranges::remove_if (C++20)
 移除满足特定条件的元素 (算法函数对象)

remove_copy

remove_copy_if
 复制范围并忽略满足特定条件的元素 (函数模板)

ranges::remove_copy (C++20)

ranges::remove_copy_if (C++20)
 复制范围并忽略满足特定条件的元素 (算法函数对象)

unique
 移除范围中连续重复元素 (函数模板)

ranges::unique (C++20)
 移除范围中连续重复元素 (算法函数对象)

unique_copy
 创建某范围的不含连续重复元素的副本 (函数模板)

ranges::unique_copy (C++20)
 创建某范围的不含连续重复元素的副本 (算法函数对象)

顺序变更操作
 在标头 `<algorithm>` 定义

reverse
 逆转范围中的元素顺序 (函数模板)

ranges::reverse (C++20)
 逆转范围中的元素顺序 (算法函数对象)

reverse_copy
 创建范围的逆向副本 (函数模板)

ranges::reverse_copy (C++20)
 创建范围的逆向副本 (算法函数对象)

rotate
 旋转范围中的元素顺序 (函数模板)

ranges::rotate (C++20)
 旋转范围中的元素顺序 (算法函数对象)

rotate_copy
 复制并旋转元素范围 (函数模板)

ranges::rotate_copy (C++20)
 复制并旋转元素范围 (算法函数对象)

shift_left (C++20)

shift_right (C++20)
 迁移范围中元素 (函数模板)

ranges::shift_left (C++23)

ranges::shift_right (C++23)
 迁移范围中元素 (算法函数对象)

`random_shuffle` (C++17 前)

`shuffle` (C++11)

随机重排范围中元素 (函数模板)

`ranges::shuffle` (C++20)

随机重排范围中元素 (算法函数对象)

采样操作

在标头 `<algorithm>` 定义

`sample` (C++17)

从序列中随机选择 N 个元素 (函数模板)

`ranges::sample` (C++20)

从序列中随机选择 N 个元素 (算法函数对象)

排序和相关操作

要求

部分算法要求由实参表示的序列“已排序”或“已划分”。未满足要求时行为未定义。

序列 `[start, finish)` 在满足以下条件时已按表达式 `f(e)` 划分: 存在一个整数 `n`, 使得对于 `[0, std::distance(start, finish))` 中的所有整数 `i`, `f(*(start + i))` 当且仅当 `i < n` 时是 `true`。

划分操作

在标头 `<algorithm>` 定义

`is_partitioned` (C++11)

判断范围是否已按给定谓词划分 (函数模板)

`ranges::is_partitioned` (C++20)

判断范围是否已按给定谓词划分 (算法函数对象)

`partition`

将范围中元素分为两组 (函数模板)

`ranges::partition` (C++20)

将范围中元素分为两组 (算法函数对象)

`partition_copy` (C++11)

复制范围并将元素分为两组 (函数模板)

`ranges::partition_copy` (C++20)

复制范围并将元素分为两组 (算法函数对象)

`stable_partition`

将元素分为两组, 同时保留其相对顺序 (函数模板)

`ranges::stable_partition` (C++20)

将元素分为两组, 同时保留其相对顺序 (算法函数对象)

`partition_point` (C++11)

定位已划分范围的划分点 (函数模板)

`ranges::partition_point` (C++20)

定位已划分范围的划分点 (算法函数对象)

排序操作

在标头 `<algorithm>` 定义

sort

将范围按升序排序 (函数模板)

ranges::sort (C++20)

将范围按升序排序 (算法函数对象)

stable_sort

将范围中元素排序, 同时保持相等元之间的顺序 (函数模板)

ranges::stable_sort (C++20)

将范围中元素排序, 同时保持相等元之间的顺序 (算法函数对象)

partial_sort

将范围中前 N 个元素排序 (函数模板)

ranges::partial_sort (C++20)

将范围中前 N 个元素排序 (算法函数对象)

partial_sort_copy

复制范围中元素并部分排序 (函数模板)

ranges::partial_sort_copy (C++20)

复制范围中元素并部分排序 (算法函数对象)

is_sorted (C++11)

检查范围是否已按升序排列 (函数模板)

ranges::is_sorted (C++20)

检查范围是否已按升序排列 (算法函数对象)

is_sorted_until (C++11)

找出最大的有序子范围 (函数模板)

ranges::is_sorted_until (C++20)

找出最大的有序子范围 (算法函数对象)

nth_element

将给定范围部分排序, 确保其按给定元素划分 (函数模板)

ranges::nth_element (C++20)

将给定范围部分排序, 确保其按给定元素划分 (算法函数对象)

二分搜索操作 (在已划分范围上)

在标头 `<algorithm>` 定义

lower_bound

返回首个不小于给定值的元素的迭代器 (函数模板)

ranges::lower_bound (C++20)

返回首个不小于给定值的元素的迭代器 (算法函数对象)

upper_bound

返回首个大于给定值的元素的迭代器 (函数模板)

ranges::upper_bound (C++20)

返回首个大于给定值的元素的迭代器 (算法函数对象)

equal_range

返回匹配特定键值的元素范围 (函数模板)

`ranges::equal_range` (C++20)

返回匹配特定键值的元素范围 (算法函数对象)

`binary_search`

判断元素是否在偏序范围中 (函数模板)

`ranges::binary_search` (C++20)

判断元素是否在偏序范围中 (算法函数对象)

集合操作 (在已排序范围上)

在标头 `<algorithm>` 定义

`includes`

当一个序列是另一个的子序列时返回 `true` (函数模板)

`ranges::includes` (C++20)

当一个序列是另一个的子序列时返回 `true` (算法函数对象)

`set_union`

计算两个集合的并集 (函数模板)

`ranges::set_union` (C++20)

计算两个集合的并集 (算法函数对象)

`set_intersection`

计算两个集合的交集 (函数模板)

`ranges::set_intersection` (C++20)

计算两个集合的交集 (算法函数对象)

`set_difference`

计算两个集合的差集 (函数模板)

`ranges::set_difference` (C++20)

计算两个集合的差集 (算法函数对象)

`set_symmetric_difference`

计算两个集合的对称差 (函数模板)

`ranges::set_symmetric_difference` (C++20)

计算两个集合的对称差 (算法函数对象)

归并操作 (在已排序范围上)

在标头 `<algorithm>` 定义

`merge`

合并两个有序范围 (函数模板)

`ranges::merge` (C++20)

合并两个有序范围 (算法函数对象)

`inplace_merge`

就地合并两个有序范围 (函数模板)

`ranges::inplace_merge` (C++20)

就地合并两个有序范围 (算法函数对象)

堆操作

(C++20 前) 随机访问范围 `[first, last)` 在满足以下条件时是一个关于比较器 `comp` 的堆: 对于 $(0, last - first)$ 中的所有整数 i , `bool(comp(first[(i - 1) / 2], first[i]))` 都是 `false`。

在标头 `<algorithm>` 定义

push_heap

添加元素到最大堆 (函数模板)

ranges::push_heap (C++20)

添加元素到最大堆 (算法函数对象)

pop_heap

移除最大堆中最大元 (函数模板)

ranges::pop_heap (C++20)

移除最大堆中最大元 (算法函数对象)

make_heap

从元素范围创建最大堆 (函数模板)

ranges::make_heap (C++20)

从元素范围创建最大堆 (算法函数对象)

sort_heap

将最大堆变成按升序排序的元素范围 (函数模板)

ranges::sort_heap (C++20)

将最大堆变成按升序排序的元素范围 (算法函数对象)

is_heap

检查给定范围是否为最大堆 (函数模板)

ranges::is_heap (C++20)

检查给定范围是否为最大堆 (算法函数对象)

is_heap_until (C++11)

查找能成为最大堆的最大子范围 (函数模板)

ranges::is_heap_until (C++20)

查找能成为最大堆的最大子范围 (算法函数对象)

最小/最大操作

在标头 `<algorithm>` 定义

max

返回给定值中较大者 (函数模板)

ranges::max (C++20)

返回给定值中较大者 (算法函数对象)

max_element

返回范围中最大元 (函数模板)

ranges::max_element (C++20)

返回范围中最大元 (算法函数对象)

min

返回给定值中较小者 (函数模板)

ranges::min (C++20)
 返回给定值中较小者 (算法函数对象)

min_element
 返回范围中最小元 (函数模板)

ranges::min_element (C++20)
 返回范围中最小元 (算法函数对象)

minmax (C++11)
 返回两个元素间的较小者和较大者 (函数模板)

ranges::minmax (C++20)
 返回两个元素间的较小者和较大者 (算法函数对象)

minmax_element (C++11)
 返回范围中的最小元和最大元 (函数模板)

ranges::minmax_element (C++20)
 返回范围中的最小元和最大元 (算法函数对象)

clamp (C++17)
 在一对边界值下夹逼一个值 (函数模板)

ranges::clamp (C++20)
 在一对边界值下夹逼一个值 (算法函数对象)

字典序比较操作
 在标头 `<algorithm>` 定义

lexicographical_compare
 当一个范围字典序小于另一个时返回 `true` (函数模板)

ranges::lexicographical_compare (C++20)
 当一个范围字典序小于另一个时返回 `true` (算法函数对象)

lexicographical_compare_three_way (C++20)
 三路比较两个范围 (函数模板)

排列操作
 在标头 `<algorithm>` 定义

next_permutation
 生成元素范围的下一个字典序更大的排列 (函数模板)

ranges::next_permutation (C++20)
 生成元素范围的下一个字典序更大的排列 (算法函数对象)

prev_permutation
 生成元素范围的下一个字典序更小的排列 (函数模板)

ranges::prev_permutation (C++20)
 生成元素范围的下一个字典序更小的排列 (算法函数对象)

is_permutation (C++11)
 判断一个序列是否为另一个序列的排列 (函数模板)

ranges::is_permutation (C++20)
 判断一个序列是否为另一个序列的排列 (算法函数对象)

数值运算

在标头 `<numeric>` 定义

`iota` (C++11)

从初始值开始连续递增填充范围 (函数模板)

`ranges::iota` (C++23)

从初始值开始连续递增填充范围 (算法函数对象)

`accumulate`

求和或折叠范围中元素 (函数模板)

`inner_product`

计算两个范围中元素的内积 (函数模板)

`adjacent_difference`

计算范围中相邻元素的差 (函数模板)

`partial_sum`

计算范围中元素的部分和 (函数模板)

`reduce` (C++17)

类似 `std::accumulate`, 但不依序执行 (函数模板)

`exclusive_scan` (C++17)

类似 `std::partial_sum`, 第 i 个和中排除第 i 个输入 (函数模板)

`inclusive_scan` (C++17)

类似 `std::partial_sum`, 第 i 个和中包含第 i 个输入 (函数模板)

`transform_reduce` (C++17)

应用可调用对象, 然后乱序规约 (函数模板)

`transform_exclusive_scan` (C++17)

应用可调用对象, 然后计算排除扫描 (函数模板)

`transform_inclusive_scan` (C++17)

应用可调用对象, 然后计算包含扫描 (函数模板)

8 杂项

8.1 二项式系数相关

二项式系数定义, 上指标可以取任意实数, 下指标取任意整数

$$\binom{r}{k} = \begin{cases} \frac{r(r-1)\cdots(r-k+1)}{k(k-1)\cdots(1)} = \frac{r^{\underline{k}}}{k!}, & \text{整数 } k \geq 0 \\ 0, & \text{整数 } k < 0 \end{cases}$$

$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad \text{整数 } n \geq k \geq 0$	阶乘展开式
$\binom{n}{k} = \binom{n}{n-k}, \quad \text{整数 } n \geq 0, k \text{ 是整数}$	对称恒等式
$\binom{r}{k} = \frac{r}{k} \binom{r-1}{k-1}, \quad \text{整数 } k \neq 0$	吸收/提取恒等式
$\binom{r}{k} = \binom{r-1}{k} + \binom{r-1}{k-1}, \quad k \text{ 是整数}$	加法/归纳恒等式
$\binom{r}{k} = (-1)^k \binom{k-r-1}{k}, \quad k \text{ 是整数}$	上指标反转
$\binom{r}{m} \binom{m}{k} = \binom{r}{k} \binom{r-k}{m-k}, \quad m, k \text{ 是整数}$	三项式恒等式
$\sum_k \binom{r}{k} x^k y^{r-k} = (x+y)^r, \quad \text{整数 } r \geq 0 \text{ 或者 } x/y < 1$	二项式定理
$\sum_{k \leq n} \binom{r+k}{k} = \binom{r+n+1}{n}, \quad n \text{ 是整数}$	平行求和法
$\sum_{0 \leq k \leq n} \binom{k}{m} = \binom{n+1}{m+1}, \quad \text{整数 } m, n \geq 0$	上指标求和法
$\sum_k \binom{r}{k} \binom{s}{n-k} = \binom{r+s}{n}, \quad n \text{ 是整数}$	范德蒙德卷积公式

8.2 顶和底相关

就放个结论

设 $f(x)$ 是任意一个具有如下性质且在一个实数区间连续的单调递增函数

$$f(x) \text{ 是整数} \implies x \text{ 是整数}$$

于是, 只要 $f(x)$ 、 $f(\lfloor x \rfloor)$ 和 $f(\lceil x \rceil)$ 都有定义, 我们就有

$$\lfloor f(x) \rfloor = \lfloor f(\lfloor x \rfloor) \rfloor \text{ 和 } \lceil f(x) \rceil = \lceil f(\lceil x \rceil) \rceil$$

其中有个重要特例: 如果 m 和 n 是整数且分母 n 为正, 则

$$\left\lfloor \frac{x+m}{n} \right\rfloor = \left\lfloor \frac{\lfloor x \rfloor + m}{n} \right\rfloor \text{ 和 } \left\lceil \frac{x+m}{n} \right\rceil = \left\lceil \frac{\lceil x \rceil + m}{n} \right\rceil$$

8.3 生成函数相关

数列	生成函数	封闭形式
$\langle 1, 0, 0, 0, 0, 0, \dots \rangle$	$\sum_{n \geq 0} [n = 0] z^n$	1
$\langle 0, \dots, 0, 1, 0, 0, \dots \rangle$	$\sum_{n \geq 0} [n = m] z^n$	z^m
$\langle 1, 1, 1, 1, 1, 1, \dots \rangle$	$\sum_{n \geq 0} z^n$	$\frac{1}{1 - z}$
$\langle 1, -1, 1, -1, 1, -1, \dots \rangle$	$\sum_{n \geq 0} (-1)^n z^n$	$\frac{1}{1 + z}$
$\langle 1, 0, 1, 0, 1, 0, \dots \rangle$	$\sum_{n \geq 0} [2 \mid n] z^n$	$\frac{1}{1 - z^2}$
$\langle 1, 0, \dots, 0, 1, 0, \dots, 0, 1, 0, \dots \rangle$	$\sum_{n \geq 0} [m \mid n] z^n$	$\frac{1}{1 - z^m}$
$\langle 1, 2, 3, 4, 5, 6, \dots \rangle$	$\sum_{n \geq 0} (n + 1) z^n$	$\frac{1}{(1 - z)^2}$
$\langle 1, 2, 4, 8, 16, 32, \dots \rangle$	$\sum_{n \geq 0} 2^n z^n$	$\frac{1}{1 - 2z}$
$\langle 1, 4, 6, 4, 1, 0, 0, \dots \rangle$	$\sum_{n \geq 0} \binom{4}{n} z^n$	$(1 + z)^4$
$\langle 1, c, \binom{c}{2}, \binom{c}{3}, \dots \rangle$	$\sum_{n \geq 0} \binom{c}{n} z^n$	$(1 + z)^c$
$\langle 1, c, \binom{c+1}{2}, \binom{c+2}{3}, \dots \rangle$	$\sum_{n \geq 0} \binom{c+n-1}{n} z^n$	$\frac{1}{(1 - z)^c}$
$\langle 1, c, c^2, c^3, \dots \rangle$	$\sum_{n \geq 0} c^n z^n$	$\frac{1}{1 - cz}$
$\langle 1, \binom{m+1}{m}, \binom{m+2}{m}, \binom{m+3}{m}, \dots \rangle$	$\sum_{n \geq 0} \binom{m+n}{m} z^n$	$\frac{1}{(1 - z)^{m+1}}$
$\langle 0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \rangle$	$\sum_{n \geq 1} \frac{1}{n} z^n$	$\ln \frac{1}{1 - z}$
$\langle 0, 1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots \rangle$	$\sum_{n \geq 1} \frac{(-1)^{n+1}}{n} z^n$	$\ln(1 + z)$
$\langle 1, 1, \frac{1}{2}, \frac{1}{6}, \frac{1}{24}, \frac{1}{120}, \dots \rangle$	$\sum_{n \geq 0} \frac{1}{n!} z^n$	e^z

8.4 Tips

- Python 中刷新缓冲区使用 `sys.stdout.flush()` (需 `import sys`)